A SCIENTIFC THESIS ON

DUAL-STAGE VAE-GAN WITH WAVELET PATCHGAN FOR ROBUST MULTI-NOISE MRI DENOISING

Submitted in partial fulfillment for the award of the degree of

Bachelor of Technology in ELECTRONICS AND COMPUTER ENGINEERING

by

ABILESH SD (22BLC1169)

VIT®
Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)
CHENNAI

SCHOOL OF ELECTRONICS ENGINEERING

April, 2026

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DECLARATION

I hereby declare that the thesis entitled "DUAL-STAGE VAE-GAN WITH WAVELET PATCHGAN FOR ROBUST MULTI-NOISE MRI DENOISING" submitted by Abilesh SD (22BLC1169) , for the award of the degree of Bachelor of Technology in Computer Science and Engineering, Vellore Institute of Technology, Chennai is a record of bonafide work carried out by me under the supervision Dr. Jani Anbarasi L.

I further declare that the work reported in this thesis has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

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Date: 07/04/26

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CERTIFICATE

This is to certify that the report entitled **"DUAL-STAGE VAE-GAN WITH WAVELET PATCHGAN FOR ROBUST MULTI-NOISE MRI DENOISING"** is prepared and submitted by **Abilesh SD** to Vellore Institute of Technology, Chennai, in partial fulfillment of the requirement for the award of the degree of **Bachelor of Technology in ELECTRONICS AND COMPUTER ENGINEERING** is a bonafide record carried out under my guidance. The project fulfills the requirements as per the regulations of this University and in my opinion meets the necessary standards for submission. The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma and the same is certified.

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ABSTRACT

Magnetic resonance imaging (MRI) involves several noise sources like Gaussian noises, Rician noises, Poisson noises, speckles, multiplicative noises and structured periodic noise that often negatively affect image quality, making it challenging to conduct proper clinical diagnosis. The additional complications of these degradations include the acquisition characteristics that are vendor specific as well as heterogeneous noise patterns that are experienced in real life clinical setting. Deep learning-based denoising methods have been extensively used to alleviate these issues to provide radiologists with better image quality and maintain structure with diagnostically significant information. Most of the available methods are, however, optimized to a single noise distribution and do not have much strength in complex and mixed noise scenarios.

Our proposal in this work is a Dual-stage variational auto-encoder/generative adversarial network (VAE-GAN) paired with a wavelet-domain patch-GAN discriminator that can be used to robustly denoise MRI. The framework proposed combines a two-step variational generator which iteratively filters noisy inputs and enforces overall anatomical regularity by bypassing a probabilistic latent bottleneck. Simultaneously, a PatchGAN discriminator that is implemented using a discrete wavelet transform works on multi-scale frequency subbands and thus allows successful retention of low-frequency structural information as well as high-frequency anatomical features. In order to assess rigorously robustness, a unified experimental pipeline is created which includes synthetic vendor-specific noise simulations of Siemens, GE and Philips scanners, all-noise combined corruption experiment, single-noise ablation experiments and Rician noise severity assessment.

Experiments on a curated glioma MRI dataset and 1,297 samples, obtained by the Kaggle BRATS collection reveal that the proposed approach is always more successful than the strong baselines like DnCNN, ResUNet and SwinIR. Quantitative data demonstrate that PSNR and SSIM are significantly improved and structural similarity increases significantly which means an excellent ability to retain anatomical fidelity. The proposed variational framework with two phases, a wavelet-domain discriminator, is accurate in tackling heterogeneous MRI noise and provides a powerful and versatile denoising framework that is applicable in a wide range of clinical imaging applications.

Keywords: MRI denoising, Dual-Stage VAE-GAN, Wavelet PatchGAN, Vendor-specific noise, Robustness, Ablation study, Glioma MRI.

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Place: Chennai

Date: 07/04/26 **Abilesh SD**

CONTENTS

LIST OF FIGURES

LIST OF TABLES

LIST OF ACRONYMS

Acronym	Full Form
VAE-GAN	Variational Autoencoder – Generative Adversarial Network
MRI	Magnetic Resonance Imaging
PSNR	Peak Signal-to-Noise Ratio
SSIM	Structural Similarity Index Measure
DWT	Discrete Wavelet Transform
VAEs	Variational Autoencoders
GANs	Generative Adversarial Networks
CNNs	Convolutional Neural Networks
DnCNN	Denoising Convolutional Neural Network
SNR	Signal-to-Noise Ratio
TV	Total Variation
BRATS	Brain Tumor Segmentation (from Kaggle BRATS collection)
GE	General Electric (scanner manufacturer)
LL	Low-Low (Wavelet Subband)
LH	Low-High (Wavelet Subband)
HL	High-Low (Wavelet Subband)
HH	High-High (Wavelet Subband)

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

Magnetic Resonance Imaging (MRI) has taken the center stage of modern medical imaging and is now commonly employed in the non-invasive diagnosis and characterization of diseases and in the planning of treatments of the neurological and oncological disorders. Despite the fact that MRI is a priceless tool in clinical practice, it is prone to noise caused by hardware constraints, movement, and accelerated scan sequences. These artifacts reduce the quality of images, cause blurring of fine anatomical structures and make it difficult to interpret the images clinically reliably. In practice, the noise distribution often distributes heterogeneously, and robust denoising is a long-standing problem in MRI due to the corruption of MRI data.

In MRI, the forms of noise include Gaussian, Rician, Poisson, speckle, multiplicative and structured periodic artifacts which have varying effects on anatomical structures at different frequencies. These artifacts may result in loss of low contrast tissue edges or high frequency distortion leading to loss of diagnostic confidence. Also, various scanner manufacturers (Siemens, GE, and Philips) use different acquisition pipelines, making the noise profile vendor-specific which creates additional challenges in general acquisition of denoising.

Denoising techniques based on deep learning have been widely employed to mitigate these problems to provide radiologists with higher quality of images without loss of diagnostically valuable information. Majority of the existing techniques are however optimized to a single noise distribution and are not robust in mixed, complex noise situations.

1.2 BACKGROUND AND MOTIVATION

Traditional MRI denoising algorithms like Gaussian filtering and wavelet thresholding are dependent on hand crafted filters which blur anatomical edges and deteriorate structural fidelity. The residual-based DnCNN model of convolutional neural networks (CNNs) has been shown to have excellent denoising performance on Gaussian noise conditions but not heterogeneous or non-Gaussian noise conditions [4], [10].

Generative models have also become strong substitutes in restoring medical images. Variational Autoencoders (VAEs) add probabilistic latent features which encourage globally consistent reconstruction and greater robustness through latent regularization [1], [8]. The Generative Adversarial Networks (GANs) with the help of adversarial training generate perceptually real images by matching reconstructed images to clean data distribution [2], [7]. GAN systems like DAGAN have been shown to be better in texture recovery in MRI denoising in tasks [6]. Nevertheless, even image-space adversarial learning might fail to capture fine-grained frequency details that can be used to retain subtle anatomical details.

The frequency-domain analysis is a complementary approach. Discrete Wavelet Transform (DWT) allows decomposing images in multi-resolution frequency subbands, hence allowing noise removal selectively without compromising structural information [3]. Recent approaches that jointly learn to use both spatial and frequency-domain representations to improve denoising have been discovered using multi-level wavelet CNNs [5]. Although these developments have been made, current models are usually only applicable to one of the three complimentary paradigms: probabilistic modeling, adversarial learning, or frequency-domain analysis, which makes them a less powerful tool in unstable, multi-source noise conditions.

1.3 PROBLEM STATEMENT

The main problem in this paper is the strong denoising of MRI images in the heterogeneous and multi-source noise conditions, which comprises the Gaussian, Rician, Poisson, speckle, multiplicative, and structured periodic noise. Current algorithms are mostly optimized over a single noise distribution and do not scale to different noise profiles encountered in real-life clinical settings, such as vendor specific acquisition pipelines of Siemens, GE and Philips scanners.

Moreover, the existing methods fail to combine probabilistic latent modeling, adversarial frequency-sensitive supervision, and step-by-step reconstruction. This is the reason to design a complex denoising architecture that can suppress heterogeneous noise adaptively without affecting diagnostically significant anatomical structures.

1.4 OBJECTIVES

The following are the main objectives of this project:

- To create a Dual-Stage Variational Autoencoder1420 Variational Generative Adversarial Network (VAE-GAN) to learn progressive denoising of MRI under heterogeneous noise.
- To design a wavelet-domain PatchGAN discriminator which is capable of frequency-sensitive adversarial supervision in order to provide better structural detail conservation.
- To test the suggested framework strictly in terms of vendor-specific noise pipelines (Siemens, GE, Philips), all-noise combined corruptions, single-noise ablations, and levels of Rician noise severity.
- To quantitatively compare the proposed model with the other existing baselines such as DnC CNN, ResUNet, and SwinIR on the basis of PSNR and SSIM.

1.5 CONTRIBUTIONS

The main contributions of this work are summarized as follows:

Dual-stage variational denoising architecture: A two-stage VAE-based generator, which sequentially denoises noisy MRI inputs, with global anatomical consistency being imposed by probabilistic latent modelling [1], [8].

Wavelet-domain adversarial learning: A PatchGAN discriminator applied to discrete wavelet transform subbands and allows the frequency-aware discrimination as well as better preservation of fine structural details [2], [3], [5].

Extensive robustness analysis: An integrated experimental pipeline, including vendor-specific noise evaluation, all-noise joint corruption, single-noise ablations, and Rician noise severity testing on a filtered glioma MRI dataset of 1,297 samples of the BRATS benchmark [9], [11].

1.6 ORGANIZATION OF THE REPORT

The rest of this report will have the following structure. Chapter 2 provides an exhaustive literature review of the available MRI denoising techniques including traditional signal processing techniques, CNN-based techniques, and generative models. The third chapter provides the detailed description of the proposed Dual-Stage VAE-GAN architecture (including the design of the generator, the loss functions and the wavelet-domain discriminator). Chapter 4 displays the experiment design, quantitative findings, baseline comparisons, and ablation experiments. Chapter 5 presents a conclusion of the report summing up the findings and providing the future work directions.

CHAPTER 2

LITERATURE REVIEW

2.1 TRADITIONAL MRI DENOISING METHODS

Before deep learning took over the field, researchers working on MRI denoising had to make do with classical signal processing techniques. The dominant approaches — Gaussian filtering, median filtering, total variation minimization, and wavelet-based thresholding — were not designed specifically for MRI.They are generic tools developed in the larger image processing community that have been used as is or "adapted" into the medical imaging area, often incompletely. Most of these tools work on an overly simplistic premise: noise occupies high frequency bands of the frequency domain, while important anatomical features lie within low frequency bands. If you can reduce the high frequency components (noise), then you will also be reducing some of the signal (anatomically important information). While this makes sense in theory; it has proven to be a gross over-simplification in practice.

For example, gaussian filtering reduces noise through the process of blurring images with a gaussian filter. As one would expect, this does eliminate much of the noise present. However, when an image is blurred, edges become less defined (i.e., tissue boundaries and lesion margins) and small structural changes may be less visible to the radiologist. The amount of blurring is determined by the size of the standard deviation of the kernel being applied. Finding an optimal standard deviation is not always easily accomplished.. Too small, and the noise remains. Too large, and diagnostically relevant details start disappearing into a blur. For routine denoising of natural photographs, this might not matter much. For MRI images where a few pixels can represent the difference between a tumor margin and healthy tissue, it matters a great deal.

Median filtering offered a somewhat better alternative for certain types of noise, particularly impulse or salt-and-pepper noise, because it replaces each pixel with the median of its neighborhood rather than a weighted average. This makes it more resistant to extreme outliers. However, it still struggles with the kinds of noise commonly found in MRI acquisitions — Rician noise, for instance, follows a non-Gaussian distribution that arises specifically from the magnitude reconstruction of complex MRI signals. Applying a method designed with Gaussian or impulse noise in mind to Rician-corrupted data is a fundamental mismatch, and the results reflect that.

The wavelet thresholding approach was clearly a more principled direction of progress. Instead of applying a uniform filtering process on the entire image in the spatial domain, wavelet techniques allow an analysis of the image through its decomposition into a number of frequency bands at various resolutions. The rationale behind the concept is that the noise, being spectrally white, tends to be distributed uniformly over all frequency bands, whereas the signal energy would be concentrated in a few larger coefficient bands. Through thresholding and nullifying the small coefficients, one can eliminate the effects of the noise while preserving the signal to a great extent.

But even wavelet thresholding has its limits. The thresholds are typically set globally or semi-locally, without any understanding of the image content. A threshold that works well for a smooth background region might be entirely wrong for a region with fine anatomical texture. Furthermore, these methods make no use of any learned prior about what brain MRI images are supposed to look like. They treat every image as an independent signal decomposition problem, with no shared knowledge across patients or scanning conditions. As clinical MRI data frequently exhibits vendor-specific characteristics — Siemens, GE, and Philips scanners each introduce slightly different noise profiles due to their respective acquisition pipelines — these fixed-parameter approaches struggle to maintain consistent performance. The result is a method that might work decently on a specific dataset under specific conditions, but generalizes poorly to the heterogeneous, multi-source noise environments found in real hospitals.

Total variation (TV) minimization methods attempted to address the over-smoothing problem by explicitly penalizing the gradient magnitude of the image, encouraging piecewise-smooth reconstructions that preserve sharp edges. TV-based denoising became popular in compressed sensing MRI reconstruction and achieved good results in settings where the noise was Gaussian and the image had a piecewise constant structure. But natural MRI images are not piecewise constant — they contain smooth intensity gradients, fine textures, and subtle variations that TV regularization tends to suppress. Staircase artifacts, where smooth gradients are replaced by blocky constant regions, are a well-known failure mode of TV methods that further reduces their clinical utility.

Looking back, the trajectory of classical denoising methods tells a consistent story. Each new technique addressed a specific limitation of its predecessor, but none was able to escape the fundamental constraint of operating without any learned knowledge of what the underlying clean signal should look like. They could suppress noise in a generic sense, but they could not distinguish between noise and diagnostically relevant fine structure because they had no model of either. That is the gap that deep learning eventually filled, though not without introducing its own complications.

2.2 DENOISING METHODS BASED ON DEEP LEARNING

The transition towards deep learning for denoising was rather slow in terms of initial research efforts, but when it happened, it was rather sudden and revolutionary. Initially, CNNs have proved themselves quite capable in solving the task by applying their learned representations to images. All the neural network needed was a set of paired training examples. The network would learn to discern noise, no matter how complex, and eliminate it without resorting to any manual feature design or filtering operations. The potential of such approach turned out to be quite large.

One of the most prominent research efforts in this field comes from Zhang et al., and it goes by the name of DnCNN. While previous methods attempted to train the network to output the actual denoised image, DnCNN employed residual learning instead, thus being able to estimate the noisy component directly. The clean image is then obtained through subtraction of the estimated noise from the input sample. It may seem as if this was a trivial change in architecture, but the advantages it brings are numerous. For one thing, the network now needs to learn mapping residuals only.

DnCNN achieved strong results on Gaussian noise with known variance and demonstrated that a single network could generalize across a range of noise levels using a technique called blind denoising. It set the template for a generation of CNN-based denoisers that followed.

The application of CNN-based methods to MRI-specific noise presented both opportunities and complications. On the opportunity side, medical imaging datasets while smaller than natural image datasets — provided paired noisy-clean data that could be used for supervised training. On the complication side, MRI noise does not behave like the additive white Gaussian noise that most CNN denoisers were designed for. Rician noise, the dominant noise model in MRI magnitude images, follows a non-central chi distribution and has a signal-dependent variance. At low SNR, it manifests as a characteristic noise floor that raises the baseline intensity of the image, particularly in background regions. Naively applying a Gaussian-trained denoiser to Rician-corrupted MRI typically results in suboptimal performance, sometimes introducing new artifacts in the process.

The scientists reacted to this issue by building MRI-specific training pipelines. The neural networks were trained using MRI images degraded by noise through the addition of Rician noise, helping the model understand the patterns associated with real MRI scans. Tian et al. (2019) reviewed various deep learning strategies for image denoising and provided an inventory of the quickly expanding area of CNN applications. The authors identified some of the most relevant architectural options in designing such neural networks. These include multi-scale structures, skip connections, and dense connectivity pattern. The authors of the study chose the U-Net architecture as a good starting point because of its ability to incorporate both low and high-level features using its skip connections.

Self-supervised approaches like Noise2Void (Krull et al., 2019) and Noise2Self (Batson and Royer, 2019) addressed a practical limitation of supervised denoising: the requirement for paired noisy-clean training data. In many clinical settings, acquiring a perfectly clean reference image alongside a noisy acquisition is either impossible or impractical. Noise2Void cleverly exploited the statistical independence of noise across pixels to train a denoising network using only single noisy images, without any clean references. The blind-spot network architecture prevents the network from simply copying the noisy pixel value, forcing it to infer the underlying signal from its surroundings instead. While this is a genuinely elegant approach, the performance ceiling is lower than that of supervised methods — the network has less information to work with, and it shows in the quantitative results.

Transformer-based architectures entered the picture more recently, with SwinIR (Liang et al., 2021) demonstrating that the self-attention mechanism of vision transformers could model long-range spatial dependencies in images more effectively than convolutions. CNNs, by their local nature, have limited receptive fields unless stacked very deep. Transformers, by attending globally across the entire image, can capture structural relationships between distant regions potentially useful in MRI denoising where the global anatomy constrains what any local region should look like. SwinIR achieved strong PSNR values on standard denoising benchmarks and has been adopted as a strong baseline in the medical imaging denoising literature.

Nevertheless, high PSNR values may not necessarily indicate clinically relevant improvements in terms of structural similarity, and the lower SSIM values reported by SwinIR under some circumstances indicate that the two are not always correlated. Despite the wide array of techniques available using deep learning, one common problem is that many such techniques are optimized only for a specific type of noise distribution or a set thereof, and their robustness to out-of-distribution noise is poor.. A network tuned for Rician noise may not handle structured periodic artifacts gracefully. This becomes a serious practical concern in clinical MRI, where different scanners, field strengths, pulse sequences, and acquisition parameters all contribute to a noise environment that is far more complex and variable than any single benchmark dataset can capture. The field has increasingly recognized this limitation, and more recent work has pushed toward architectures that can handle heterogeneous, multi-source noise conditions which is precisely the motivation behind the proposed framework.

2.3 GENERATIVE ADVERSARIAL NETWORKS FOR MEDICAL IMAGE RESTORATION

The Generative Adversarial Network framework (GAN), invented by Goodfellow et al., introduces a radically new perspective into image generation and restoration. At its heart, GANs represent a clever hack: rather than train one network to generate output images that optimize some predefined loss function, one trains two networks in tandem, pitting them against one another.In particular, one network attempts to create realistic images, while the other network (the discriminator) aims to distinguish whether a generated image is fake or not. The outcome of this competition between the two networks is such that both networks end up learning how to imitate the statistical characteristics of the data distribution in question, resulting in output that looks like actual images from the data distribution in question, rather than trying to minimize mean square errors in order to recreate the actual image.

This approach may sound pedantic, but it is crucially important. Optimization with respect to a pixel-level loss function, such as mean squared error, tends to encourage the generator to produce the average of all possible solutions that can be obtained for a particular input. This is problematic in the case of corrupted images, because such inputs typically do not

DAGAN, proposed by Chen et al. (2024), is one of the more widely cited GAN-based denoisers in the medical imaging literature. It demonstrated convincingly that adversarial training could preserve tissue textures in MRI in ways that CNN baselines could not. Perceptual quality metrics, as well as qualitative visual inspection by radiologists, consistently favored the GAN-generated outputs. The key advantage was in fine-grained structural detail things like the texture of white matter, the sharpness of sulcal boundaries, and the appearance of small lesion margins that regression-based methods tended to smooth over. These are exactly the kinds of features that matter clinically, and the improvement was tangible.

However, standard GAN-based denoisers operate in the image domain, and this comes with a subtle but important limitation. The discriminator in a standard GAN sees the entire image as a flat two-dimensional array of pixel values. It has no built-in mechanism to reason about frequency content or to distinguish between low-frequency structural information and high-frequency textural detail. In case the generator is able to create an image which looks globally plausible with regard to global anatomical structures as well as global intensity distribution, the discriminator could be misled even when the fine-level high-frequency features are wrong or missing. The use of patch-based discriminators, known as PatchGAN, which have been used in image-to-image translation tasks, partly solve this problem by considering image quality on a patch-wise basis instead of a global one. However, even PatchGAN doesn't specifically consider image decomposition based on frequencies.

The DISGAN algorithm introduced by Wang et al. (2023) is a very important milestone in terms of frequency-oriented GANs. While PatchGAN uses patches directly from images, DISGAN starts by applying Discrete Wavelet Transform to the input images and then trains the discriminator using subbands obtained after that.

This design choice has a clear theoretical motivation: different frequency subbands capture different aspects of image quality. The low-frequency LL subband captures global structure and anatomy, while the high-frequency subbands (LH, HL, HH) capture edges, textures, and fine details. By discriminating in all of these spaces simultaneously, the adversarial signal is much more informative about where the generated image deviates from the real one in terms of frequency content. DISGAN showed measurable improvements in structural preservation on MRI super-resolution and denoising tasks, validating the value of frequency-domain adversarial supervision.

The QWD-GAN framework by Yang et al. (2025) took another step towards generalization of wavelet-guided GAN frameworks, using unsupervised image-to-image translation in denoising of medical microscopy images, showing that the same advantages of frequency-aware training remain valid even without MRI applications. This is an additional indication that wavelet-informed discriminators are not beneficial due to any specific properties of the problem domain but because the ability to analyze frequencies allows the discriminative component to provide better feedback to the generative model.

In addition, it should be mentioned that there is a wide range of works on stabilization of GAN training that might prove useful for our work. As is known, GANs are one of the most unstable deep learning architectures, prone to such problems as mode collapse, vanishing gradients, and discriminator overpowering. The issue is further complicated in medical imaging where datasets tend to be small and overfitting is always possible. Different solutions were proposed to ensure stable GAN training, including such measures as spectral normalization, gradient penalty term, progressive growing of networks, and proper learning rate scheduling of generator and discriminator models. The proposed Dual-Stage VAE-GAN framework will include some of these stability strategies, such as fixation of the discriminator

2.4 VARIATIONAL AUTOENCODERS FOR PROBABILISTIC MODELING OF IMAGES

VAEs, which were presented in 2013 by Kingma and Welling, and then improved upon by Doersch (2016), are situated within a slightly different niche in the world of generative models when compared to GANs. While the latter learn indirectly by competing against each other in an adversarial game, the former learn directly via maximizing a lower bound of the log-probability of data. The fundamental difference of VAEs compared to other autoencoder architectures is that VAEs define a probabilistic latent space: while a conventional autoencoder maps its input into a deterministic code vector, a VAE encoder learns a set of parameters of a probability distribution of a latent representation, usually a Gaussian one, while a latent vector itself is sampled from this distribution using reparameterization trick.

KL-divergence term in the VAE objective works as a regularizer on the latent space, bringing the learned distribution close to a standard normal prior distribution.

This regularization has an important practical consequence: it prevents the network from simply memorizing training examples by encoding each one to a distinct, isolated region of latent space.On the other hand, it promotes a smooth and continuous latent manifold wherein semantically similar images will have close points in the latent space. As such, for the generation of images, the process becomes quite straightforward in that when a sample is taken from the prior, we would expect a good looking image to be generated from it. In image reconstruction applications, such as denoising, the latent representation should encode information about the structure of the image in a compact and generalizable way.

Such probabilistic nature of bottleneck is especially handy in the setting of denoising task. Since the deterministic autoencoder is trained to generate predictions for each pixel on top of the noisy data, the network has to come up with certain values for every one of them. If the image is really uncertain and noise levels are high, there might be no point estimating exact pixel value – all possible clean image configurations should be considered. A variational autoencoder, however, encodes the very uncertainty in the distribution of latent space points. The distribution itself depends on the level of uncertainty – i.e., how unsure the network was in reconstructing the latent vector based on a noisy image.

Aetesam and Maji (2024) demonstrated these advantages concretely in the context of MRI-specific Rician noise. Their conditional variational network learned to condition the latent distribution on the noisy input, producing denoised outputs that outperformed several deterministic baselines on standard MRI benchmarks. The improvements were particularly noticeable at higher noise levels, which is precisely the regime where deterministic networks suffer most from their inability to represent uncertainty. The probabilistic framework allowed the network to remain confident in high-SNR regions while appropriately hedging in regions where noise corruption was more severe.

VAEs are not without their own limitations, of course. The reconstruction quality of a vanilla VAE is often lower than that of a GAN the images can look slightly blurry compared to the sharp outputs of a well-trained adversarial model. It is due to the inherent nature of the pixel-wise reconstruction loss in conjunction with the KL term that VAEs usually use during training. While minimizing MSE between true and reconstructed image pixels even with stochastic sampling of the intermediate layer, it forces the network to do the reconstruction in terms of the means. One reason why hybrid approaches that incorporate both VAE and adversarial elements have been successful is their complementary goals.

In particular, VAEs provide regularization of the latent representation, whereas the adversarial component focuses on generating high-fidelity perceptually realistic images.

Dual-stage architecture of the proposed approach generalizes the classic VAE bottleneck into a two-stage process.The first-stage VAE focuses on coarse reconstruction suppressing the dominant noise components and restoring the global anatomical layout. Then, the second-stage VAE acts upon the intermediate reconstruction together with the original input. The former is utilized to extract residual noise and recover finer details that were not captured by the first-stage VAE due to smoothing and approximation artifacts. In this regard, the rationale for using a two-stage framework relies on the intuition that the optimization procedure associated with recovering coarse structure while reconstructing fine-grain details at the same time is highly unstable and difficult to train.

From a theoretical perspective, the proposed two-stage approach can be interpreted as a hierarchical latent space. While the first-stage VAE captures coarse anatomical priors of the MRI, the second-stage VAE extracts fine residual structure. Hierarchical VAEs have been extensively studied in the generative modeling community to model multi-scale structure within complicated distributions. Similarly, the two-stage process proposed in this paper leverages hierarchical latent spaces to address the multi-scale nature of clean images by recovering coarse anatomical structure and fine-grain texture details from a noisy input signal.

2.5 WAVELET-DOMAIN METHODS IN DEEP LEARNING

There have been many applications of the Discrete Wavelet Transform to signal and image processing prior to deep learning by a number of decades. This method gained popularity due to the ability of wavelets to break down a signal into scale-oriented components. When applied to images, the wavelet decomposition technique helps to find an optimal trade-off between low-frequency information and high-frequency noise. In this respect, noise and signal behave differently in the domain of wavelets. Namely, being spectrally broadband and spatially uncorrelated, noise tends to distribute itself evenly among all the sub-bands. Signal is a more correlated component of an image and therefore contains most of its energy in a relatively small subset of coefficients of a larger magnitude.

The initial motivation behind integrating wavelet decomposition with deep neural networks was to gain more computational efficiency. Unlike the traditional approach where information loss happens due to max-pooling operation, using wavelets to downsample does not lose any information; it only rearranges information into sub-bands. This ensures that the network processes the images on a low-resolution scale, while all the information related to the frequency domain is preserved in the other sub-bands to be retrieved at some point in time. This feature is especially relevant when dealing with applications where the focus lies on restoring fine detail structures.

One such application was addressed by MWCNN architecture introduced by Liu et al. in 2018. The key insight was to replace standard pooling and upsampling operations with DWT and inverse DWT respectively, allowing the network to perform all of its computation in the wavelet domain. The subbands from different decomposition levels are concatenated and processed jointly, enabling the network to simultaneously leverage information from different frequency scales. MWCNN achieved competitive results on standard image restoration benchmarks and demonstrated that spatial and frequency-domain representations could be combined within a single end-to-end trainable architecture.

The extension of wavelet-domain processing to the discriminator in adversarial frameworks represents a more recent and particularly fruitful direction. In the case where the discriminator uses raw pixels as input, any feedback from the discriminator to the generator will be inherently frequency-wise entangled. The generator will never have an idea whether it failed because it got the structure right but lacked texture or vice versa. The same applies to the failure being attributed to noise at a certain frequency range. This problem is solved by training a discriminator on wavelet subbands, whereby each subband provides feedback on a specific frequency range. Thus, the LL subband provides feedback on low-frequency structures, the LH on horizontal high frequencies, the HL on vertical high frequencies, and the HH on diagonal high frequencies. This decomposed adversarial signal is substantially more informative and allows the generator to target its improvements more precisely.

DISGAN demonstrated the practical value of this approach in MRI restoration. By training the discriminator on DWT subbands rather than raw image patches, the adversarial training process became more sensitive to frequency-specific artifacts including the kinds of structured periodic noise that are particularly problematic in MRI acquisitions and notoriously difficult to remove with purely spatial methods. Periodic noise has a very specific frequency signature: it manifests as concentrated energy in particular high-frequency subbands, depending on the period and orientation of the artifact. A spatial discriminator may never explicitly detect this, because the artifact is visually subtle and spatially extended. A wavelet-domain discriminator, however, will see the anomalous energy concentration in the affected subbands immediately, providing a clear training signal to the generator to suppress it.

Hybrid approaches combining wavelet processing with other deep learning components have also shown promise.In their paper, Zangana & Mustafa (2024) developed a hybrid model that combines wavelets and CNNs for the denoising of medical images. Their study revealed superior performance in removing structured noise from images through the use of the hybrid wavelet-CNN architecture. In particular, their architecture utilizes DWT in order to first decompose medical images into subbands, followed by applying unique CNN models on different subbands, and later combining the output using IDWT. Through this approach, each branch was able to focus on the specific noise type found within each subband.

It is also worth considering what wavelet decomposition offers that learned frequency representations do not. Convolutional networks, in principle, can learn arbitrary frequency-selective filters from data, and one might argue that a sufficiently deep and wide CNN should be able to learn wavelet-like decompositions automatically if they are useful. In practice, however, there are good reasons to prefer explicit DWT over learned frequency decomposition. The DWT is a fixed, mathematically well-understood transform with known frequency localization properties. It requires no additional parameters to learn, introduces no additional regularization challenges, and provides guarantees about the completeness and orthogonality of the decomposition that learned filters cannot. For a discriminator, these properties are particularly valuable you want the discriminator to be a reliable, consistent judge of frequency-domain quality, not a learned approximation whose behavior may shift during training.

Taken together, the literature on wavelet-domain methods in deep learning makes a compelling case that explicit frequency decomposition is not merely a useful add-on but an architecturally important component for robust image restoration in settings with structured or heterogeneous noise. The proposed Dual-Stage VAE-GAN builds directly on this body of work by incorporating a wavelet-domain PatchGAN discriminator that operates on all four DWT subbands simultaneously, ensuring that adversarial supervision covers both low-frequency anatomical structure and high-frequency textural detail throughout the training process.

CHAPTER 3

PROPOSED METHODOLOGY

3.1 OVERVIEW

This chapter presents a detailed description of the proposed Dual-Stage Variational Autoencoder – Generative Adversarial Network (VAE-GAN) framework designed for robust MRI image denoising under heterogeneous and multi-source noise conditions. The architecture is proposed to address the limitations of the existing single-stage denoising models, which are usually trained to minimize a single noise distribution and not generalise to mixed and complex noises clinical scenarios. The suggested methodology applies three supplementary paradigms, namely probabilistic latent modeling with variational autoencoders, adversarial learning using a wavelet-domain PatchGAN discriminator, and progressive two-stage reconstruction to facilitate coarse-to-fine noise suppression. The framework is likened to the six various kinds of noises generally encountered in clinical acquisition of MRI which comprise of the Gaussian noise, Rician noise, Poisson noise, speckle noise, multiplicative noise, and structured periodic noise. These noise sources are combined with well developed vendor specific vendor noise simulation pipelines which depict the nature of acquisition of Siemens, GE and Philips MRI scanners. The proposed system is capable of dealing with all-noise mixed corruption, single-noise ablation experiments hence an all-encompassing and generalizable denoising system.

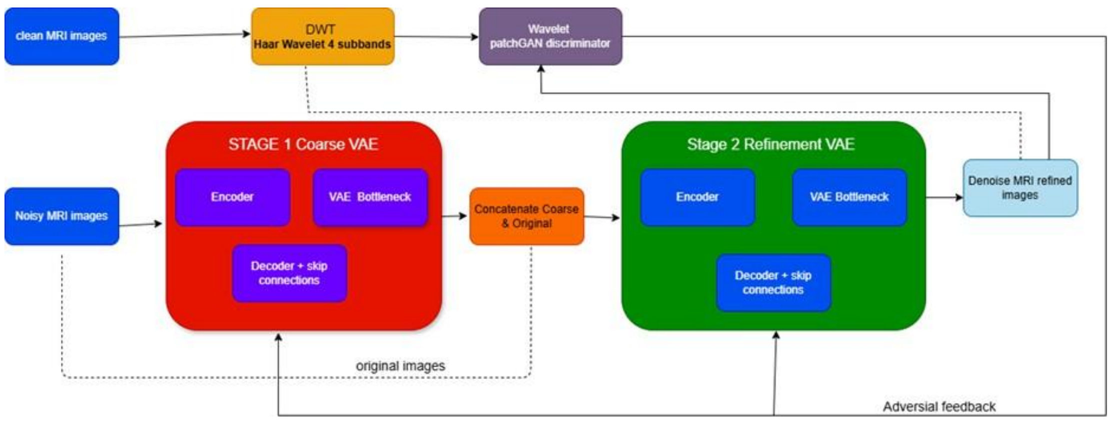


Figure 3.1: Overall system architecture of the proposed Dual-Stage VAE-GAN MRI denoising framework.

3.2 DATASET DESCRIPTION AND PREPROCESSING

The proposed framework is. Evaluated on a special set of brain MRI images. These images are from people with a type of brain tumor called glioma. The images come from the Kaggle BRATS collection. The dataset has 1,297 grayscale MRI slices of the brain. Each slice has a matching clean reference image. All these images are from patients with glioma. This makes the dataset very useful for testing how well a model can remove noise from brain MRI images. The images are prepared in the way before training, validation and testing. Each image is made to be 256 × 256 pixels. The pixel brightness is adjusted to be between 0 and 1. These steps are done using the library. They are applied to both images and noisy images. No extra techniques like rotation or flipping are used. This ensures that the results only show how well the model removes noise. The dataset is split into two parts: training and validation. The noisy and clean images are matched by filename. This ensures that they are correctly paired during training. The PyTorch DataLoader is set up with a batch size of 2 for training. The batch size is 1, for validation. The data is transferred to the GPU quickly.

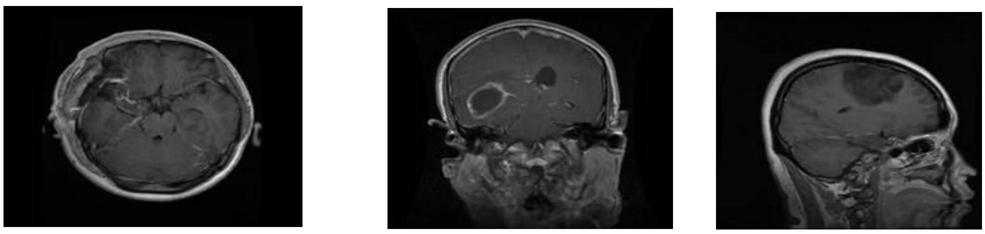


Figure 3.2: Representative clean brain MRI samples from the curated glioma dataset used for training and evaluation.

3.3 NOISE SIMULATION PROTOCOLS

A central contribution of this work is the design of a comprehensive and clinically motivated noise simulation framework. Since acquiring paired noisy-clean MRI data directly from clinical scanners is rarely feasible due to patient safety and scanner access constraints, synthetic noise generation is employed to create controlled, reproducible, and varied training conditions. The simulation protocols are structured into four categories: vendor-specific noise pipelines, an all-noise combined pipeline, single-noise ablation pipelines, and an independent Rician severity evaluation pipeline.

Individual Noise Types

Six distinct noise types are synthesized and applied independently or in combination to clean MRI images. Gaussian noise, the most commonly studied noise type in image denoising, is added as zero-mean additive white noise with a standard deviation of $\sigma = 0.05$, modeled as: $x\_noisy = clip(x + N(0, \sigma^2))$. Rician noise, which arises from the magnitude reconstruction of complex-valued MRI k-space data, is simulated by adding independent Gaussian noise to both the real and imaginary channels of the MRI signal and computing the resultant magnitude, evaluated at severity levels $\sigma \in \{0.03, 0.05, 0.07\}$. Poisson noise is modeled as signal-dependent shot noise proportional to the local image intensity, applied by scaling pixel values, generating Poisson-distributed samples, and normalizing back. Speckle noise is introduced as multiplicative noise defined by: $x\_noisy = clip(x + x \cdot N(0, \sigma^2))$, simulating granular interference patterns. Multiplicative noise is applied by element-wise multiplication with a noise field drawn from a uniform distribution: $x\_noisy = clip(x \cdot U(1-\sigma, 1+\sigma))$. Periodic structured noise is introduced as sinusoidal intensity modulation overlaid across the image to simulate scanner-induced coherent artifacts.

Vendor-Specific Noise Pipelines

Three vendor-specific noise pipelines are designed to model the acquisition characteristics of major clinical MRI scanner manufacturers. The Siemens pipeline applies a sequential combination of Gaussian noise, Rician noise, and periodic structured artifacts, reflecting the frequency of noise profiles associated with Siemens MAGNETOM systems. The GE pipeline introduces speckle noise, multiplicative intensity fluctuations, and Poisson shot noise, emulating acquisition degradations reported in GE SIGNA systems. The Philips pipeline combines Gaussian and Rician noise degradations, consistent with Philips Ingenia scanner noise characteristics. These pipelines are applied sequentially with pixel-value clipping to [1e-6, 1.0] after each noise application to prevent numerical overflow.

All-Noise Combined Pipeline

To simulate extreme worst-case degradation scenarios, an all-noise combined pipeline is constructed by sequentially applying all six noise types — Gaussian, Rician, Poisson, speckle, multiplicative, and periodic — to each clean image in a single pass. This pipeline serves as a stress-test for the denoising framework, evaluating its robustness under highly compounded noise conditions that simulate severe clinical imaging degradation. The outputs of this pipeline represent the most challenging evaluation scenario considered in this work.

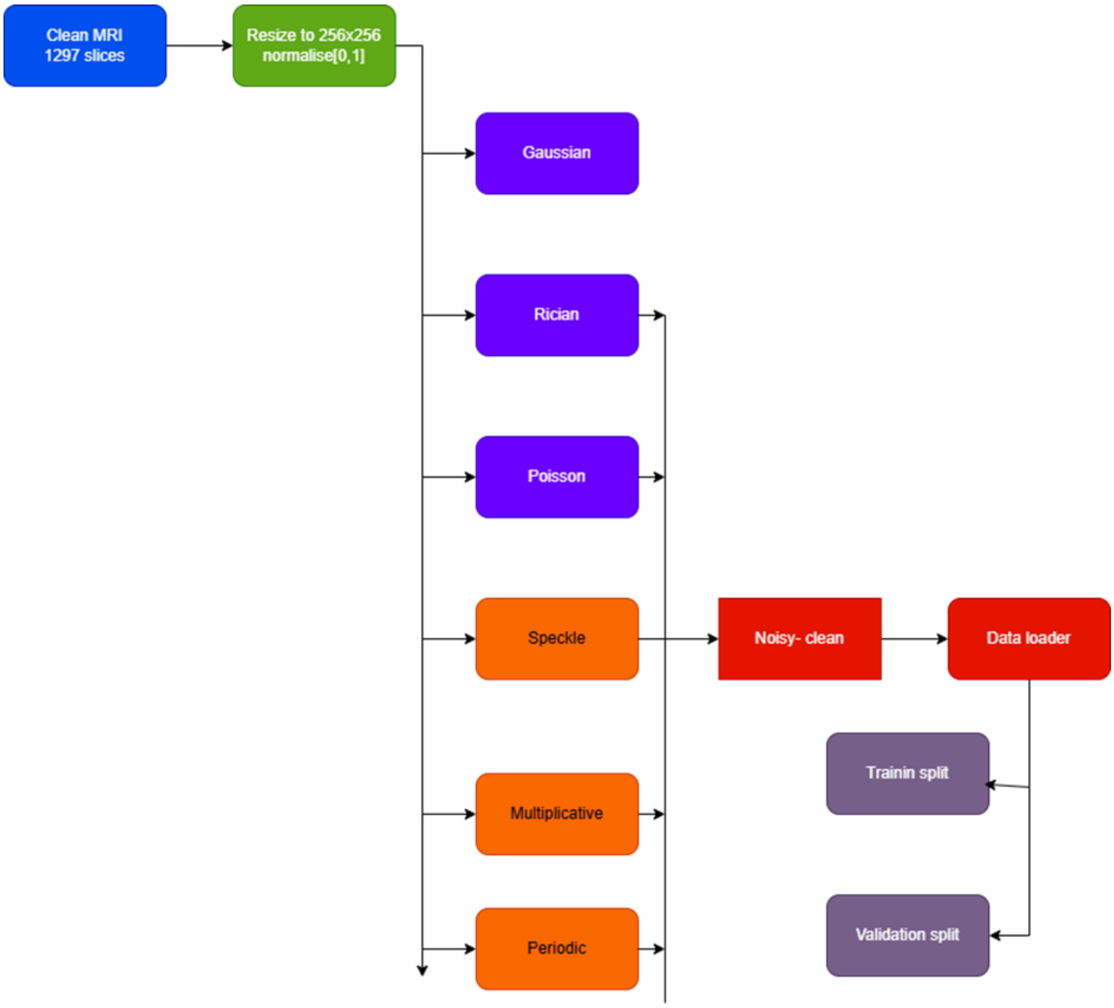


Figure 3.3: Data preprocessing and multi-noise simulation pipeline used to generate training and evaluation datasets.

3.4 MODEL ARCHITECTURE

Dual-Stage Variational Autoencoder Generator

The generator is a dual-stage variational encoder-decoder network that performs progressive denoising through two sequentially coupled variational autoencoder modules. Each stage implements a U-Net-style encoder-decoder architecture with skip connections, augmented by a probabilistic variational bottleneck that samples latent representations through the reparameterization trick. Using convolutional blocks with Instance Normalisation and LeakyReLU(0.2) activation, the architecture is built in PyTorch.

In order to avoid huge activations that may become problematic, LayerNorm normalization technique is applied to the flattened representation before it is fed through the variational linear layers to avoid so-called gradient explosion. After that, the representation goes through two independent dense layers that project it to two different representations: latent mean – which is similar to the average, and log-variance – which represents variance, in 256 dimensions each.

In order to sample from the representation of the hidden model without breaking gradient flow, the VAE uses a neat mathematical trick known as the reparameterization trick. As opposed to assigning the specific value to z (and thus losing our ability to do back-propagation), the VAE randomly samples a "noise" vector e from the Gaussian distribution and assigns $z = \mu + \sigma \otimes e$. Thus, the randomness part is isolated from those parts of the equation we wish to learn on, making us able to compute gradients for μ and σ in the process of training. In order to avoid instabilities, log-variance outputs are capped within -20..20 range before being converted into σ; preventing extremely large or small variances.
Structure of Model: Each part of the Encoder consists of four layers which uses Convolution to perform operations on data. As the data passes through the layers, the number of channels in the layers keeps increasing (input → 64 → 128 → 256 → 512). After passing through the layers, operation of MaxPool2D reduces the size of the data by half in the vertical and horizontal directions. At the end of all layers, the data will be 16x16x512. This data is then flattened into a vector of size 131,072 and passed through the fully connected layers for calculating the mean and log variance of latent variable distribution. Input is passed to the network which has undergone several sequential processes with each performing operations in a certain way. The aim is to transform the input into a certain form which can be manipulated further. Use of convolutions and MaxPool2D helps in extracting information while reducing the volume of data to be processed.

The final result is a vector that contains a lot of information, which is then used to produce the desired output.

Coarse Reconstruction

Stage 1 takes the noisy MRI input $x \in \mathbb{R}^{1\times256\times256}$ and learns a probabilistic latent representation that encodes the main noise-free anatomical structure. This is called coarse denoising. The variational encoder uses the reparameterization trick to figure out the posterior parameters (μ_1, σ_1^2) and sample the latent variable z_1. The Stage 1 decoder makes an intermediate denoised output $\hat{x}_1 = \text{Decoder}_1(z_1)$ that shows the overall shape of the anatomy while getting rid of the loudest noise parts. To keep the reconstructed pixel values between 0 and 1, a sigmoid activation is applied to the output of the last convolutional layer.

$$z_1 = \mu_1 + \sigma_1 \odot \varepsilon,\ \varepsilon \sim N(0, I) \quad ...(3.1)$$

$$\hat{x}_1 = Decoder_1(z_1) \quad ...(3.2)$$

Stage 2: Fine Refinement

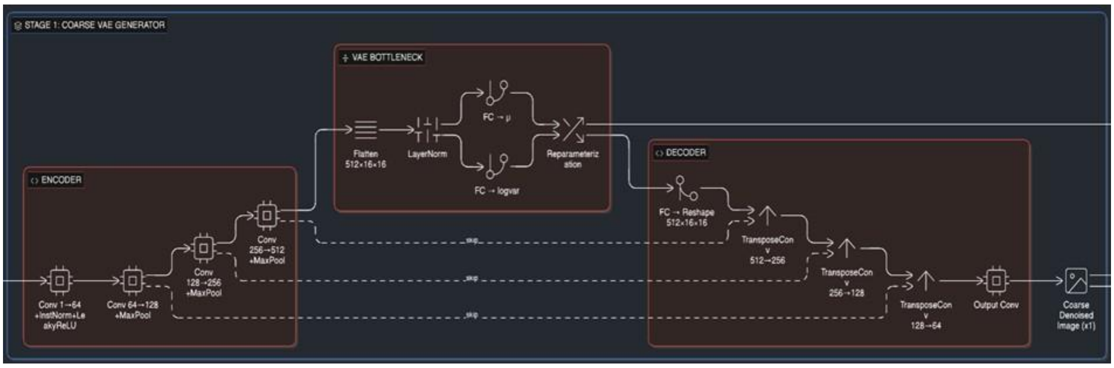
The second stage processes the two inputs to produce another input. This other input will be a combination of the previously processed (partially cleaned) image and the original noisy image. Combining these two images produces an additional layer of data which includes two components, each containing 256 sub-components. Each component contains 256 sub-components as well. Processing this combined image allows the second-stage encoder to process both the partially cleaned image and the noisy image simultaneously. This enables the encoder to remove remaining noise from the partially cleaned image, thereby further improving its quality. The second-stage encoder and decoder are designed to specifically enhance the quality of the image. They determine many of the critical aspects of detail such as; how much remaining noise exists in the image, how confident they are in their determination of detail in the image and use that information to create a cleaner version of the image. That cleaner version represents the results of processing through the second-stage clean-image process.

$z_2 = \mu_2 + \sigma_2 \odot \varepsilon$ …(3.3) $\hat{x}_2$

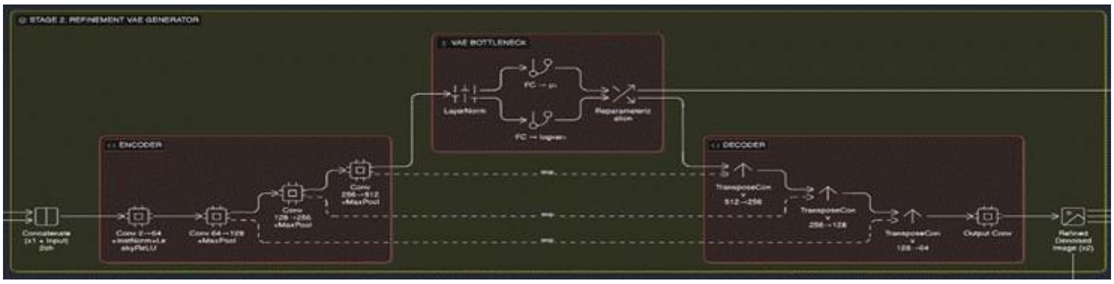
$= \text{Decoder}_2(z_2)$ …(3.4)

The reasoning behind the two-stage progressive design comes down to a fundamental limitation of single-bottleneck VAEs — asking one variational layer to simultaneously handle broad anatomical structure and subtle high-frequency noise is simply too much to optimise cleanly in one shot. The two objectives pull the network in different directions during training, which tends to produce unstable gradients and mediocre reconstructions across both tasks.

Breaking the problem into two dedicated stages sidesteps this tension. The first stage performs most of the heavy lifting, providing a broad structural framework for the image -- removing all the global noise present within the image while also establishing a basic anatomical layout based upon the raw scan data. Once a solid base has been established by the first stage, the second stage becomes focused primarily upon refining those areas of the anatomical layout that were damaged by the residual noise produced during the initial stage of processing. Because finer anatomical details require high frequency data to properly reconstruct, the second stage is able to recover those details with greater accuracy than would be possible if the details had to be reconstructed along with the broader anatomical features. Since each stage has a specific task or goal, the training requirements are significantly easier to manage, and thus do not require that the network perform multiple conflicting tasks (i.e., coarse vs. fine level of reconstruction).



Stage 1



Stage 2

Figure 3.4: Detailed architecture of the Dual-Stage Variational Autoencoder generator with skip connections and variational bottlenecks

Wavelet PatchGAN Discriminator

The discriminator doesn't look at the individual pixel values. Instead, it works with frequencies. To do this, it first passes the real clean images and the generator's denoised outputs through a special layer. This layer is called a 2D Discrete Wavelet Transform, and it uses something called Haar filter banks. The reason for using this layer is to keep everything running smoothly on the computer's graphics card, and to make sure the gradients - which are important for training the model - stay clean and easy to work with. This way, the discriminator can make better decisions about what's real and what's not.

The DWT breaks down each single-channel image into four parts, called frequency subbands, which are half the size of the original image. The LL part shows the overall shape of the image, while the LH and HL parts show the horizontal and vertical lines, and the HH part shows the diagonal details and any noise. These four parts are combined into a single unit, called a tensor, which has a size of (4, H/2, W/2) and is used as input for the PatchGAN discriminator. For the Haar filter used, the low pass filter will be $h = [1, 1]/\sqrt{2}$ and the high-pass filter will be $g = [1, -1]/\sqrt{2}$ and the downsampling is performed using a stride-2 convolution. This procedure aims at separating the image parts and preparing them for more advanced analysis.

The architecture of the discriminator has five layers based on convolution operations. They consist of 4, 64, 128, 256, 512, and 1 channel(s), respectively. The size of the kernels applied in all layers equals 4×4. Intermediate layers perform instance normalization and use activation with the LeakyReLU function of 0.2 values. In addition, the final layer changes the way patches are being scanned from larger to smaller ones. It also produces a map representing probabilities of patches being real or fake. Such an approach, referred to as PatchGAN, provides a penalty for any local inconsistency of generated images, making the generator create more realistic outputs that can be locally consistent.

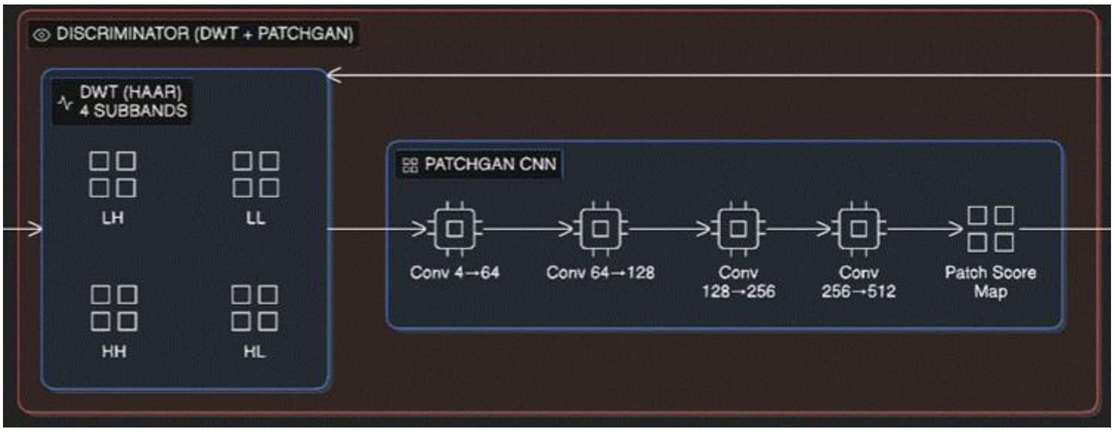


Figure 3.5: Architecture of the Wavelet-domain PatchGAN discriminator showing DWT subband decomposition and PatchGAN convolutional layers.

3.5 LOSS FUNCTIONS AND TRAINING OBJECTIVE

Generator Loss

Training was performed using the L_VAE loss function that combines the pixel-level detail quality check; the structural similarity of the entire output image compared to the input image; and the preservation of underlying logic patterns as follows:

$$L_{VAE} = \lambda_{L1}\Big(\|x - \hat{x}_1\|_1 + \|x - \hat{x}_2\|_1\Big) + \lambda_{SSIM}\Big((1 - SSIM(x, \hat{x}_1)) + (1 - SSIM(x, \hat{x}_2))\Big) + \lambda_{KL}\Big(KL_1 + KL_2\Big)$$

Given that there are several parameters involved in the weighting of each component of L_VAE, care should be taken when selecting these values. Specifically, we give a very high priority to the pixel reconstruction component by setting a weight value of 5.0. We set a lower weight value of 2.0 to the structural similarity component as well, indicating that although the model will place some importance on ensuring that the structural similarities of images are preserved, the primary focus will always be on reconstructing images.

The Kullback-Leibler Divergence component is treated differently. During training, it begins at an initial weight of zero and is incremented until it reaches a maximum of 1 × 10^(-4). By approximately 10 epochs, this incremental approach allows the model to initially concentrate on the task of reconstruction and then gradually add regularization, ultimately allowing the model to converge to more robust representations.

Backpropagation requires the use of clipped log variance values between -4.0 and +4.0 for Kullback-Leibler (KL) Divergence calculations as well as clipped output values between -10.0 and +10.0 to avoid numerical overflow. SSIM Loss instead utilizes an 11x11 fixed Gaussian window ensuring that the data and device types are properly aligned. What's interesting about the structural similarity index is that it's composed of three key components: luminance, contrast, and structural comparison.

This makes it extremely sensitive to any degradation in an image that affects how we perceive it. As a result, it's really good at picking up on changes that are important to the human eye.

Loss of the Discriminator

When you train your Discriminator you are using a type of loss called Binary Cross Entropy with Logits Loss. In addition, you are also using label smoothing. Label Smoothing means that during the time the discriminator is evaluating your images, all real images have a target of .9, and all fake images have a target of .1. As such, there is less confidence of the discriminator in being able to tell what is real and what is not. This keeps the discriminator on its toes, particularly at the beginning of training. Using a loss function to evaluate how well the discriminator does at distinguishing real and fake images is an example of what the discriminator loss function would look like.

$$L_D = \frac{1}{2}\left(BCE(D_{real}, 0.9) + BCE(D_{fake}, 0.1)\right) \quad \text{...(3.6)}$$

Generator Loss Function

To make the Denoising Output of the Generator look like a real clean MRI Image so the discriminator thinks it is, we use the Adversarial Part of the Generator Loss Function. To calculate the Generator Loss Function based on the Adversarial portion do the following:

$$L_{GAN} = BCE(D_{final}, 1.0) \quad \text{...(3.7)}$$

Combining Training Goals

In order to combine both the Variational Reconstruction Loss and the Adversarial Generator Loss into one final training goal, we add them together by combining them with a fixed value of lambda for the Adversarial Weight equal to 0.5.

$$L_{total} = L_{VAE} + \lambda_{GAN} \cdot L_{GAN} \quad \text{... (3.8)}$$

The Fixed Value for Lambda represents the amount of feedback received by the generator due to the discriminator. It was determined that a value of 0.5 was sufficient to provide a good trade-off between reconstructing an image accurately and perceiving the reconstructed image realistically.

3.6 TRAINING SETUP AND HYPERPARAMETERS

The proposed Dual-Stage VAE-GAN architecture was implemented in PyTorch and trained on a CUDA-enabled system utilizing a GPU for accelerated computation.

All convolutional, transposed convolutional, and linear layer weights were initialized using Kaiming Normal initialization, while all bias terms were set to zero. For both VAE stages, the log-variance bias was initialized to −2.0, enabling the model to begin with conservative variance estimates. To further stabilize training and prevent excessively large initial Kullback–Leibler (KL) divergence values, the weights of all variational linear layers were initialized from a narrow normal distribution, N(0, 0.001).

Optimization was performed using the Adam optimizer with parameters β_1 = 0.5 and β_2 = 0.999, applied to both the generator and the discriminator.

Training was conducted over 30 iterations (epochs), with each iteration consisting of two mini-batches. To improve training stability, the discriminator was frozen during the first three iterations, allowing the generator to learn a reasonable initialization for denoising before engaging in adversarial training.

To ensure reproducibility, random seeds were fixed to 42 across multiple libraries, including PyTorch, NumPy, Python's built-in random module, and the cuDNN backend.

Finally, prior to full-scale training, a preliminary validation step was performed to verify that the generator produced valid outputs, ensuring the absence of numerical instabilities such as NaN (Not a Number) or Inf (Infinity) values.

Table 3.1: Summary of Hyperparameters and Training Configuration

Hyperparameter / Configuration	**Value**
Input Image Size	256 × 256 (Grayscale)
Latent Dimension	256
Batch Size	2 (Train), 1 (Validation)
Training Epochs	30
Generator Learning Rate	1×10^{-4}
Discriminator Learning Rate	1×10^{-6}
Optimizer (Generator & Discriminator)	Adam (β_1=0.5, β_2=0.999)

LR Scheduler	ReduceLROnPlateau (factor=0.5, patience=5)
λ_L1 (Pixel Reconstruction Weight)	5.0
λ_SSIM (Structural Similarity Weight)	10.0
λ_KL (KL Divergence Weight)	$0 \rightarrow 1\times10^{-4}$ (10-epoch warmup)
λ_GAN (Adversarial Weight)	0.5
Discriminator Freeze Epochs	First 3 epochs
Log-Variance Clamp Range	[−4.0, 4.0]
DWT Filter	Haar (normalized)
Framework	PyTorch (CUDA)

3.7 WORKFLOW

To start with, we take clean MRI images from a dataset of glioma and make them all the same size, 256×256, and adjust the intensity so it's between 0 and 1. Next, we add synthetic noise to these images using a specific method - it could be based on the vendor, combined, or single noise - to create pairs of noisy and clean images. Then, we use a special class called MRIDenoisingDataset to load these pairs and feed them to the DataLoader in small, mixed-up groups during the training process.

Here's how the whole thing goes down. We first take a bunch of noisier-than-average input data and run that data through a specific type of encoder known as a Variational Encoder. That produces us two important things: mu1 and log sigma_1^2. Then, we use a technique known as Reparameterization Sampling to create for us a brand-new value, z1; z1 is used to create a cleaner (but still imperfect) version of our noisy input data, which we'll call x_hat_1. Next, we will mix together the partially-cleaned x_hat_1 with the original noisy input data and then we run that mixed-data through yet another Variational Encoder to obtain for us mu2 and log sigma_2^2. Finally, using Reparameterization Sampling again to generate z2, we create our final cleaned-up output, which we'll call x_hat_2. From there, we apply a unique transformation to convert both our clean real images and our generated cleaned-up versions into four frequency-based subbands. As previously mentioned, this is accomplished by applying the Discrete Wavelet Transform (DWT Layer), so that when the Discriminator evaluates these four subband representations, it will be able to distinguish the real from the fake. The Discriminator is trained to identify whether the subband representations are real or if they have been created. On the other hand, the Generator is trained to minimize a type of loss function referred to as Composite Loss L_total. Training occurs in alternating steps, where the Discriminator is trained one step at a time followed immediately by the Generator being trained the very next step. Following every single round of training, also referred to as an Epoch, we measure and record two key performance metrics — PSNR and SSIM — on a separate set of Validation Data. These metrics help determine which of the many versions of the Generator performed the best during a given Epoch. Therefore, we continue to train the entire system over and over again until such point that it has produced enough high-quality cleaned-up output to meet its intended purpose.

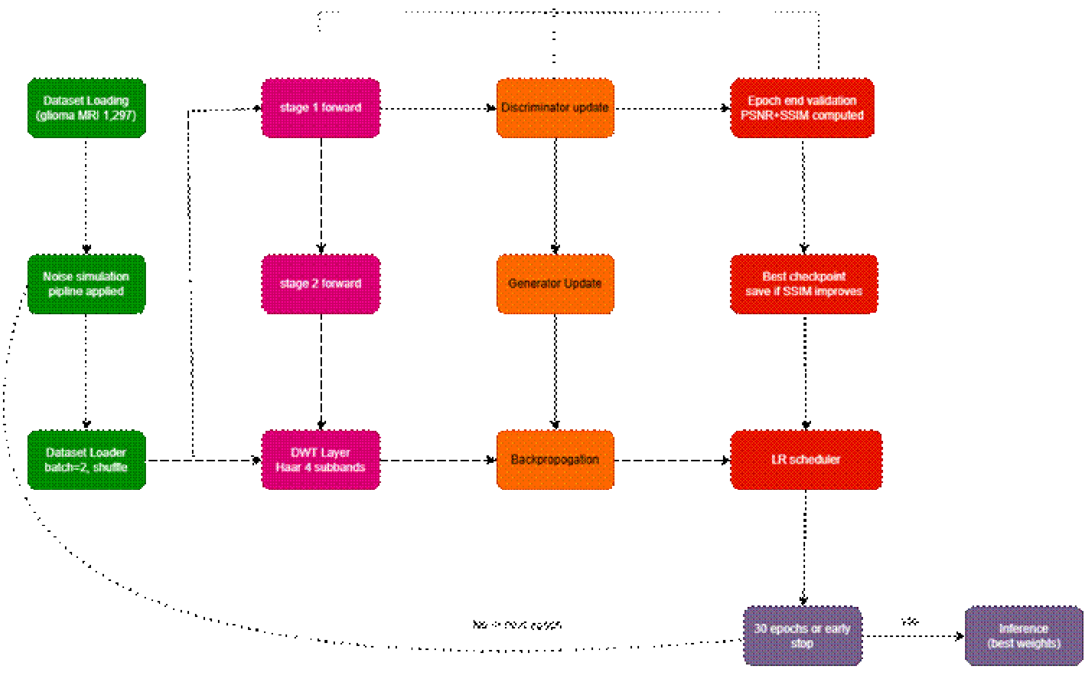


Figure 3.6: Step-by-step training workflow of the proposed Dual-Stage VAE-GAN framework.

CHAPTER 4

RESULTS AND DISCUSSION

4.1 EXPERIMENTAL SETUP

In this chapter, the experiments are performed according to a standardized and repeatable evaluation procedure to allow a reasonable comparison of various noise conditions and baseline models. The suggested Dual-Stage VAE-GAN architecture is developed in PyTorch and is trained on the NVIDIA GPU with CUDA support. The training setup is kept constant throughout all the experimental pipelines with hyperparameters as in Chapter 3, Table 3.1. Every noise pipeline produces an independent set of noisy-clean image pairs on the same base glioma MRI dataset of 1,297 samples, so that the variation in performance between pipelines is indicative of noise difficulty and not data variability.

All components are fixed to random seeds with 42 to achieve full reproducibility of training and evaluation. The most successful epoch in terms of validation SSIM saves model checkpoints, and inference is done on the best-epoch checkpoints across all reported results. The evaluation metrics are calculated on the entire validation split without any post processing on model outputs.

4.2 EVALUATION METRICS

Quantitative evaluation of the denoising quality of the framework is done with two complementary measures, namely Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM). These are the most commonly used metrics in the literature of the MRI denoising task and collectively represent the pixel-wise fidelity and the perceptual structural quality of the denoised results.

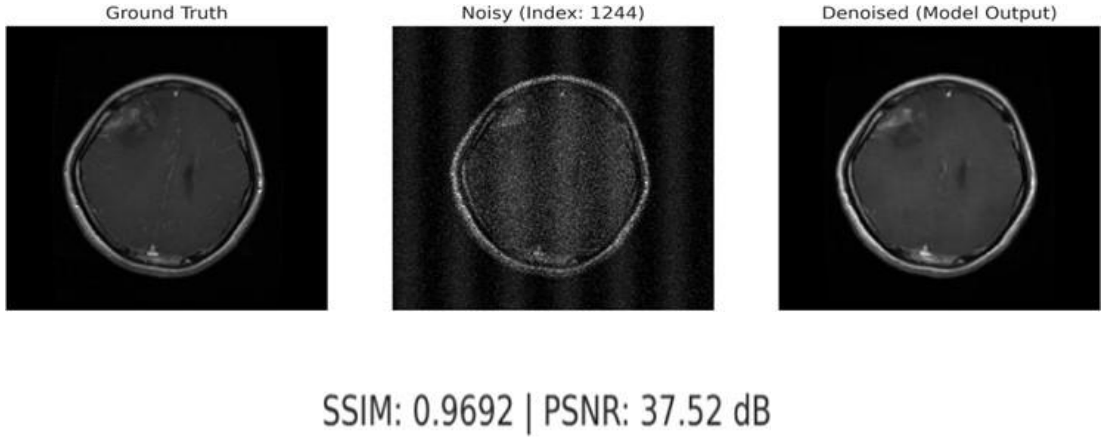
PSNR is used to measure the ratio of the maximum signal power to the power of reconstruction error in decibels. It can be expressed as PSNR = 10 log 0 (MAX 2 / MSE) where MAX = 1.0 when the images are normalized and MSE is the mean square error of the denoised image and the clean reference. The larger the PSNR the smaller the reconstruction error and the larger the signal fidelity. The scikit-image library is used to compute PSNR, and the data range is 1.0.

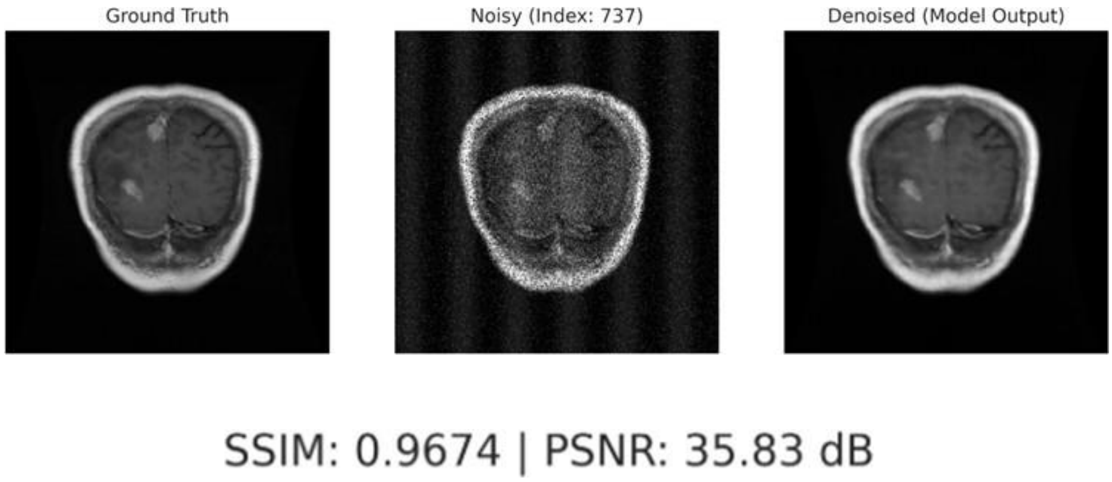
SSIM is a measure of perceptual image quality, which quantifies structural similarity between three components, namely, luminance, contrast, and structural correlation. SSIM values are in the range of 0 to 1, where a value of 1 means maximum structural identity with the reference. SSIM is especially informative in a medical imaging context since it punishes structural distortions - e.g. blurring of tissue boundaries or loss of lesion margins - more directly than simple pixel-level metrics. A large SSIM is essential in clinical practice so that the anatomical features that are important to diagnostics are not suppressed by the denoised image. SSIM is calculated with the help of scikit-image library and data-range =1.0.

4.3 QUANTITATIVE PERFORMANCE ANALYSIS

The quantitative results of the suggested Dual-Stage VAE-GAN framework are measured on all the experimental pipelines. The proposed model has a mean PSNR of 33.50 dB and a mean SSIM of 0.9423 with standard deviations of 1.35 dB and 0.0129 respectively on the vendor-specific noise pipeline. These findings indicate good noise suppression with a wide range of vendor specific noises profiles, and structural quality has been maintained.

In particular, the SSIM performance is clinically relevant. Although PSNR measures the entire reconstruction accuracy, the SSIM directly measures the maintenance of structural details used by radiologists to make a diagnosis. The proposed model attains SSIM values that are always above 0.92 on the vendor pipeline, which represents faithful preservation of anatomical structures such as tissue boundaries, sulcal patterns and lesion boundaries even with realistic compound noise conditions.





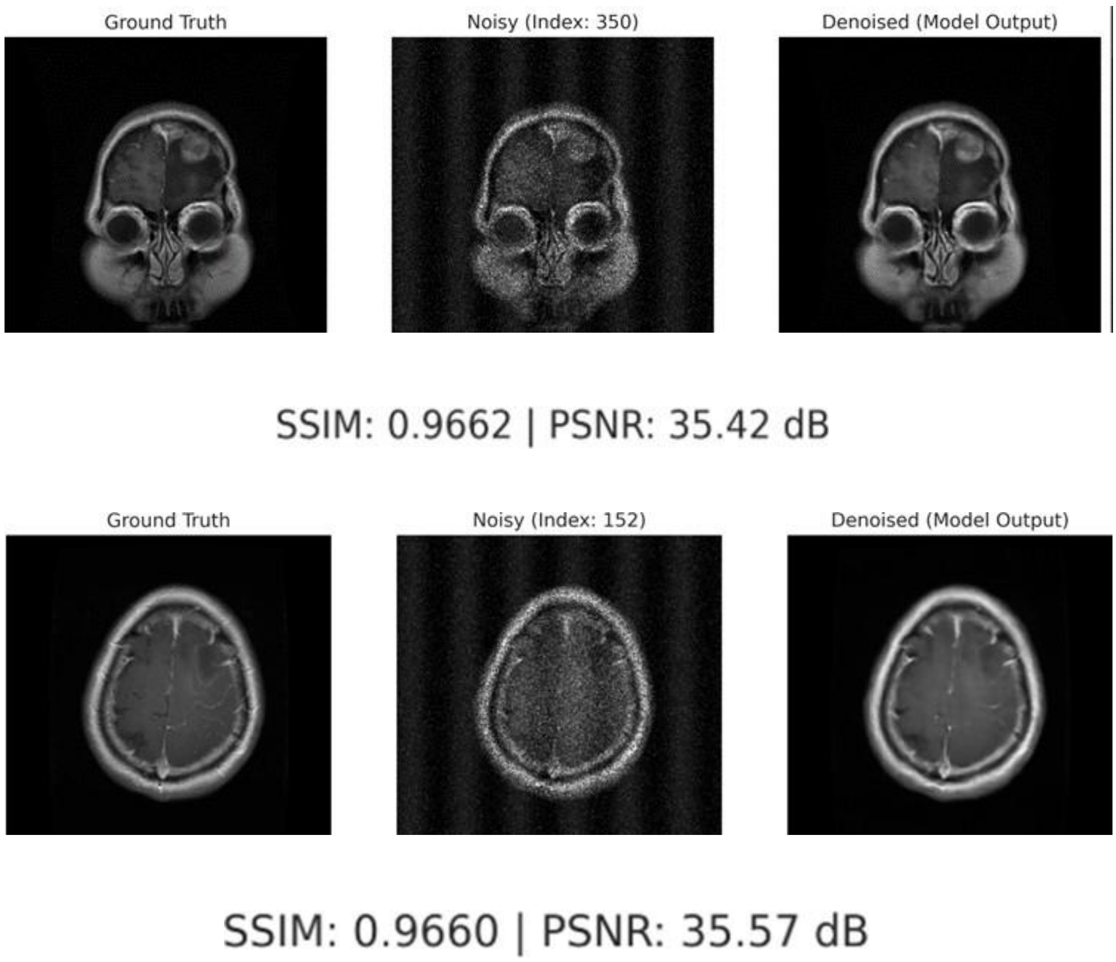


Figure 4.1: Qualitative comparison of noisy input, denoised output of the proposed Dual-Stage VAE-GAN, and clean ground truth MRI for the vendor-specific noise pipeline.

4.4 COMPARISON WITH BASELINE MODELS

In order to strictly overcontextualize the results of the proposed framework, comparative experiments are carried out with three known denoising baselines: DnCNN, ResUNet, and SwinIR. DnCNN is a residual CNN-based denoiser that predicts the noise map and not the clean image, which is the typical supervised CNN denoising method. ResUNet is a skip-connected residual U-Net architecture, which is the state of the art encoder-decoder denoising architecture in medical imaging. SwinIR is a transformer-based image restoration network that uses shifted-window self-attention, which is the most powerful transformer-based baseline in the literature. Each of the baselines is assessed in the same experimental conditions with the synthetic vendor-aware noise pipeline.

Table 4.1: Quantitative Comparison of Denoising Performance on the Vendor-Aware Noise Pipeline

Model	Mean PSNR (dB)	Mean SSIM	Std PSNR (dB)	Std SSIM
Proposed Dual-Stage VAE-GAN	**33.50**	**0.9423**	1.352228	0.012943
DnCNN	31.0353	0.7295	1.6083	0.0881
ResUNet	29.4049	0.7170	3.6080	0.0917
SwinIR	32.8725	0.8648	1.5791	0.0291

The proposed Dual-Stage VAE-GAN has the best mean PSNR of 33.50 dB and mean SSIM with 0.9423 of all the models considered. It achieves higher results on both metrics despite SwinIR being a significantly larger transformer-based architecture. The SSIM difference is more notable where the proposed model outperforms SwinIR by 0.0775 SSIM points which in medical imaging is a significant difference in the amount of structure evident in an image. SwinIR achieves 32.87 dB PSNR and its SSIM of 0.8648 implies that its global self-attention mechanism compromises structural fidelity with pixel-level accuracy in the presence of heterogeneous noise conditions.

The basic constraint of pure residual CNN-based denoising is evident in the mean SSIM of DnCNN, which is only 0.7295 when it works with the compound, vendor-specific noise distributions outside the single-noise training conditions. ResUNet has the lowest PSNR of 29.40 dB and SSIM of 0.7170 and the largest PSNR standard deviation of 3.61 dB, which means that it does not consistently denoise across different noise profiles. Both DnCNN and ResUNet report high SSIM standard deviation (0.0881 and 0.0917, respectively), which also proves their inconsistency in the case of heterogeneous noise.

Contrarily, the suggested structure has a low SSIM standard deviation of 0.0199, which means that it has a strong and stable structural preservation, regardless of various noise conditions. This regularity is explained by the fact that the variational probabilistic bottleneck and the wavelet-domain adversarial supervision are designed complementary, and they are both used to impose the global anatomical regularity and the frequency-specific structural detail preservation across the entire training process.

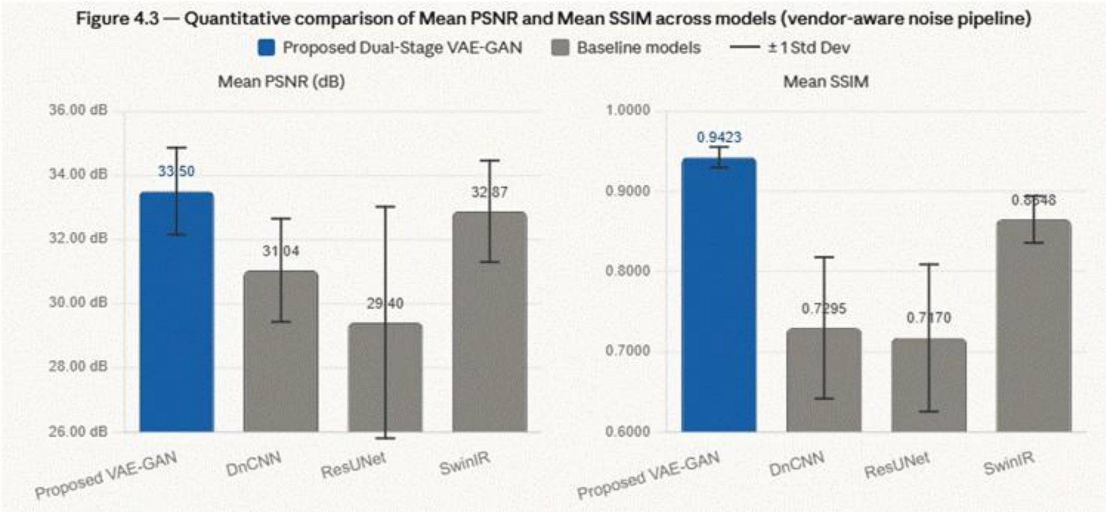


Figure 4.2: Quantitative comparison of Mean PSNR and Mean SSIM across the proposed model and baseline models on the vendor-aware noise pipeline.

4.5 ABLATION STUDY

4.5.1 Noise-Type Ablation

In order to examine the denoising robustness against the patterns of degradation on individual noise types, a thorough noise-type ablation study is performed by training and testing the proposed framework on each type of noise separately. This research shows the response of the architecture to particular noise properties without the confounding effects of noise-compound interactions. Table 4.2 shows the results of this study

.

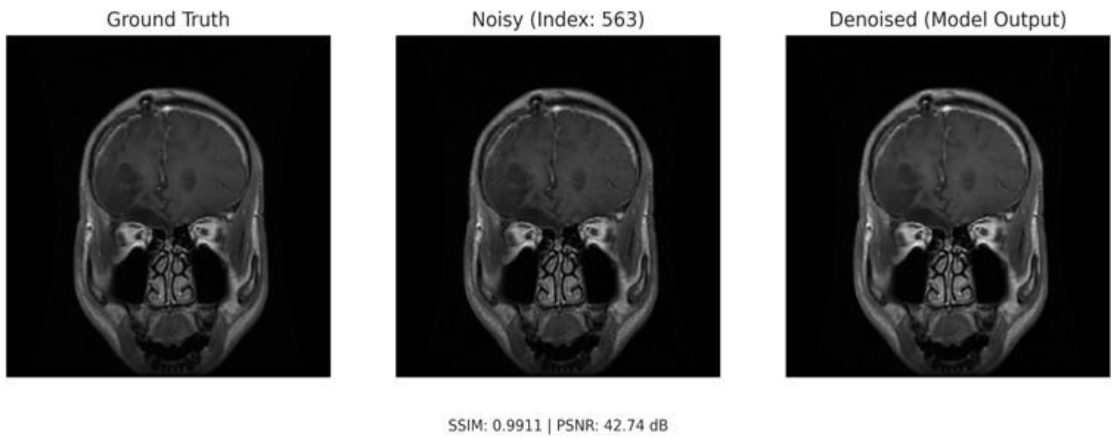
Table 4.2: Denoising Performance Across Individual Noise-Type Ablation Pipelines

Noise Pipeline	PSNR (dB)	SSIM
Rician-only (σ = 0.03)	37.52	0.9624
Rician-only (σ = 0.05)	36.41	0.9572
Rician-only (σ = 0.07)	35.05	0.9542
Gaussian-only	36.08	0.9570
Poisson-only	38.09	0.9701
Periodic-only	44.34	0.9908
Speckle-only	35.37	0.9592
Multiplicative-only	40.88	0.9853
All-Noise Combined	**33.50**	**0.9423**

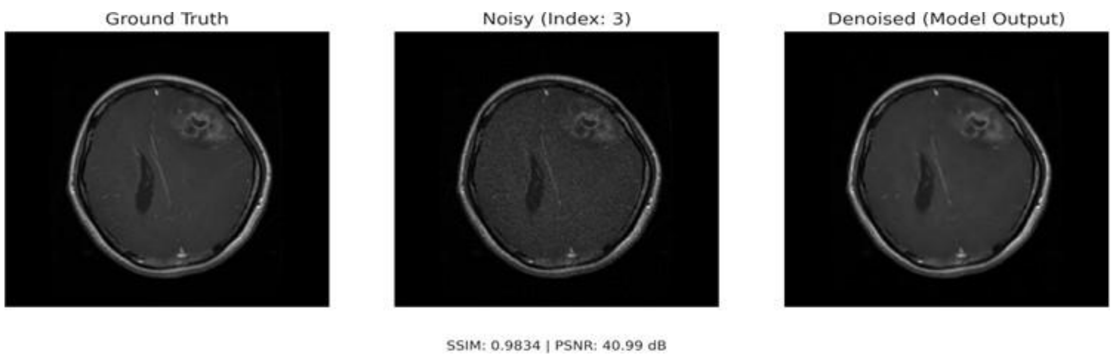
The findings of the noise-type ablation show that there are a few significant observations regarding the denoising properties of the model. The proposed structure works well with low severity (σ = 0.03) Rician noise (PSNR = 37.52 dB, SSIM = 0.9624) as this is the cleanest single-noise scenario. With a gradual increase in Rician noise severity, a steady and progressive decrease in both measures is found, which confirms that the model does not have discontinuous failure modes but increases in response to noise severity. At the maximum level of severity (0.07), the SSIM is still over 0.92, which reflects strong structural preservation at all levels.

The noise that has the best PSNR of 44.34 dB and SSIM 0.9908 occurs in periodic-only noise and all of the evaluated conditions. This finding can be directly explained by the fact that the wavelet-domain PatchGAN discriminator breaks down images into the following subbands: LL, LH, HL and HH, and uses adversarial feedback in the high-frequency components where periodic interference is located. Practically this provides the discriminator with a far sharper view of sinusoidal artifacts than a spatial-domain discriminator would have. Multiplicative noise also works well with a PSNR of 40.88 dB and SSIM of 0.9853 that indicates the ability of the VAE bottleneck to capture signal-dependent losses by using probabilistic latent sampling instead of deterministic feature extraction.

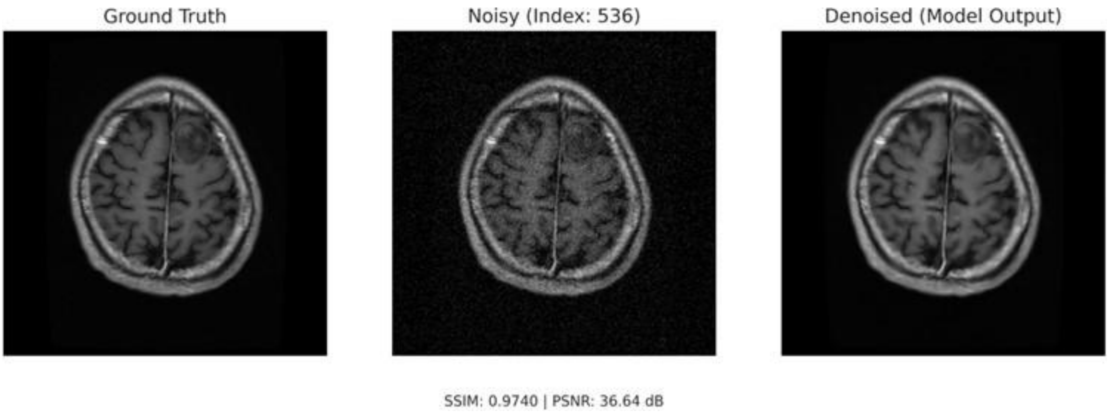
The most difficult condition is the all-noise combined pipeline, which has a PSNR of 33.50 dB and SSIM of 0.9423. Although the corruption is severely compounded with all six types of simultaneous noise types compounded in sequence, the model has an SSIM of 0.94, which is high and shows that the model preserves its structure well even when the noise is extreme. This finding confirms the original objective of the dual-stage progressive denoising architecture, which removes noise of compounds one step at a time, instead of trying to reduce noise effects in a single reconstruction step.



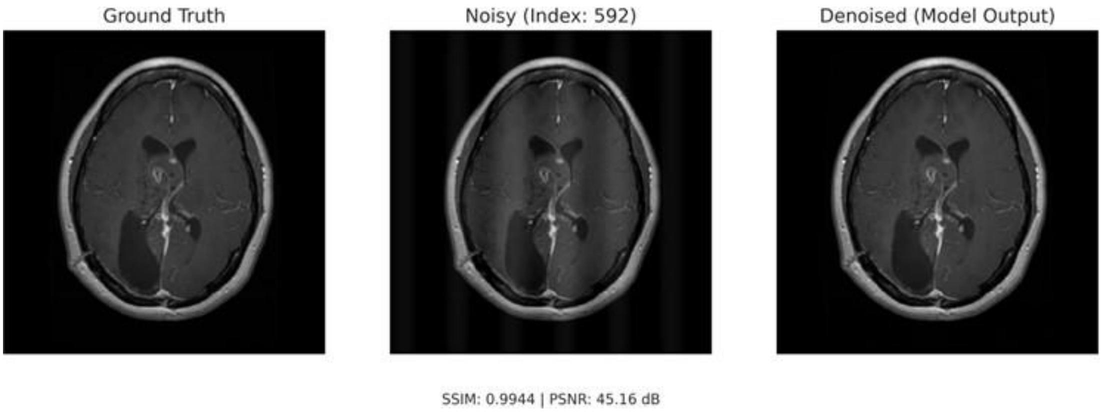
Multiplicative Noise



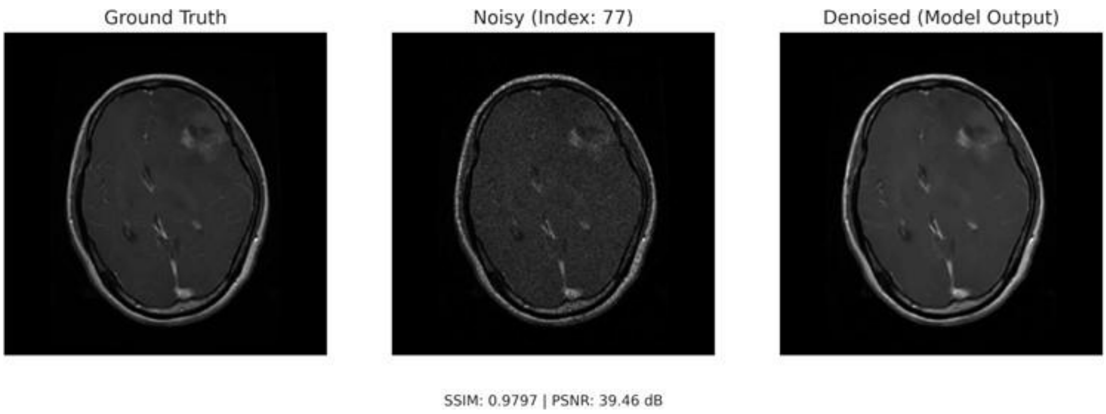
Poisson noise



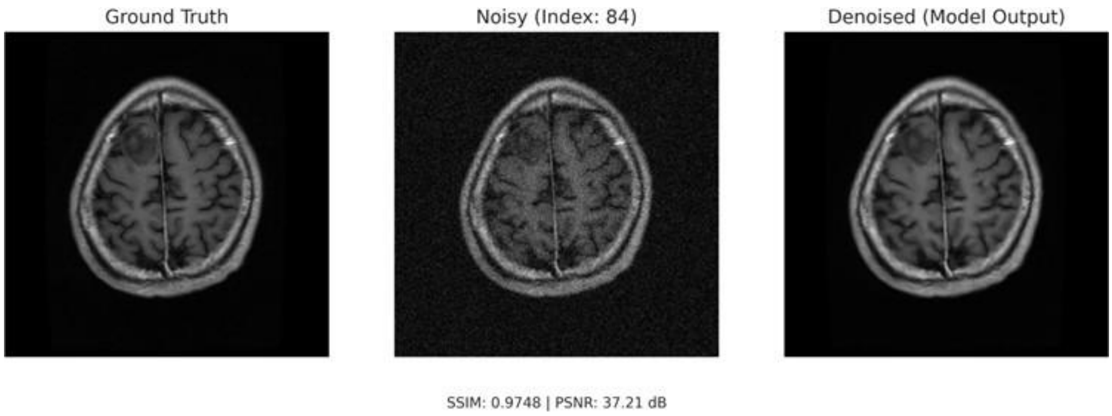
Gaussian noise



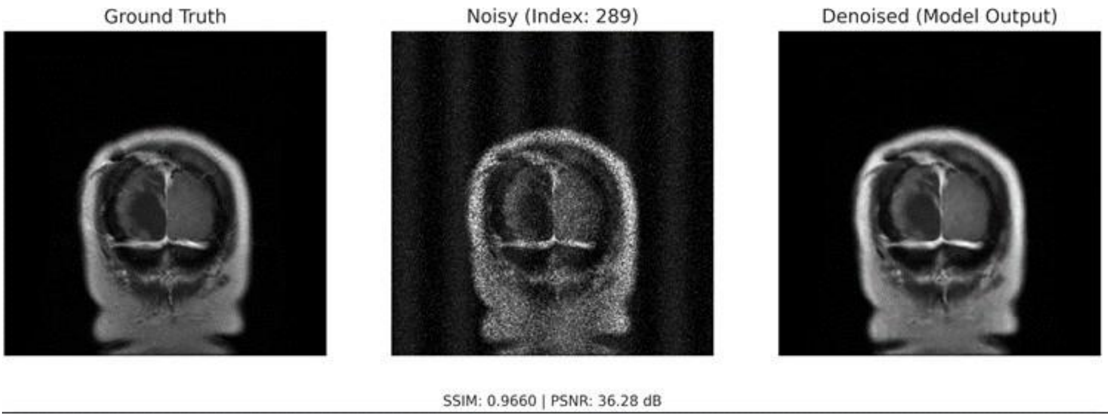
Periodic noise



Speckle noise



Rician noise(0.05)



All Noise combined

Figure 4.3: Qualitative denoising results across individual noise types, demonstrating the model's robustness under diverse degradation patterns.

4.5.2 Rician Noise Severity Analysis

In order to further analyse the model behaviour in the context of a clinically dominant noise type a dedicated Rician noise severity analysis is performed by training and testing the proposed framework on its own at three severity levels of noise: 0.03 (low severity), 0.05 (moderate severity) and 0.07 (high severity). Rician noise is a noise distribution that is produced in magnitude reconstruction of complex-valued MRI k-space data and is the most realistic noise distribution in clinical MRI acquisition. The direct diagnostic importance of the theoretical understanding of how the model scales with Rician severity is thus of direct diagnostic interest.

The inferences made out of Table 4.2 indicate an obvious and monotonic decline in PSNR and SSIM with the increase in the noise level between 0.03 and 0.07. The model has a minimum severity of 37.52 dB PSNR and 0.9624 SSIM that indicates a fairly mild noise level at which the VAE bottleneck is capable of reliably restoring anatomical structure with only a small amount of ambiguity. At moderate severity (σ = 0.05) performance settles at 36.41 dB and 0.9572 and at high severity (σ = 0.07) the model yields 35.05 dB and 0.9542. Surprisingly the SSIM drop in the entire range of severity is a narrow range of only 0.0082. This implies that the framework does not deteriorate in a harsh manner but instead smoothly deteriorates as noise level increases and critically, ensures that SSIM remains above 0.95 even in the presence of high levels of Rician corruption. This development is shown in Figure 4.6 both on the metrics.

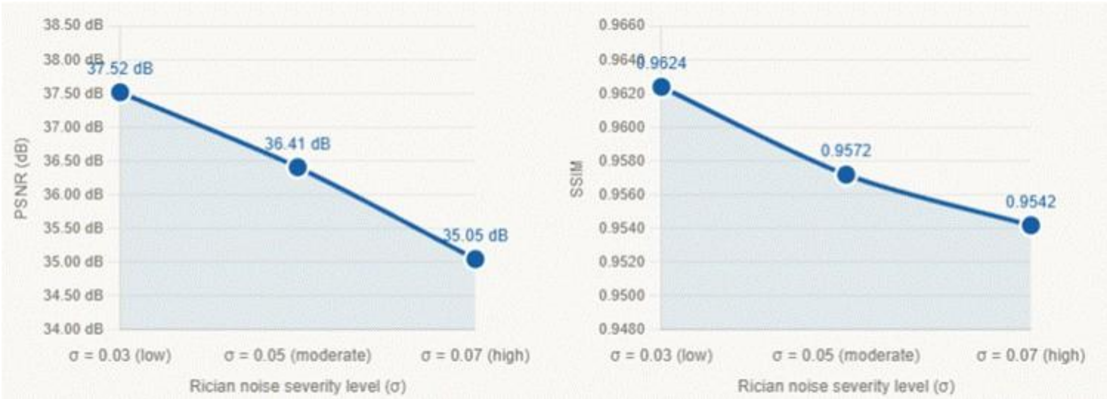


Figure 4.4: Performance variation of the proposed Dual-Stage VAE-GAN across Rician noise severity levels (σ = 0.03, 0.05, 0.07). Both PSNR and SSIM decrease monotonically with increasing noise severity. SSIM remains above 0.95 across all tested levels indicating clinically acceptable structural fidelity throughout the evaluated Rician severity range.

4.5.3 Architectural Component Ablation

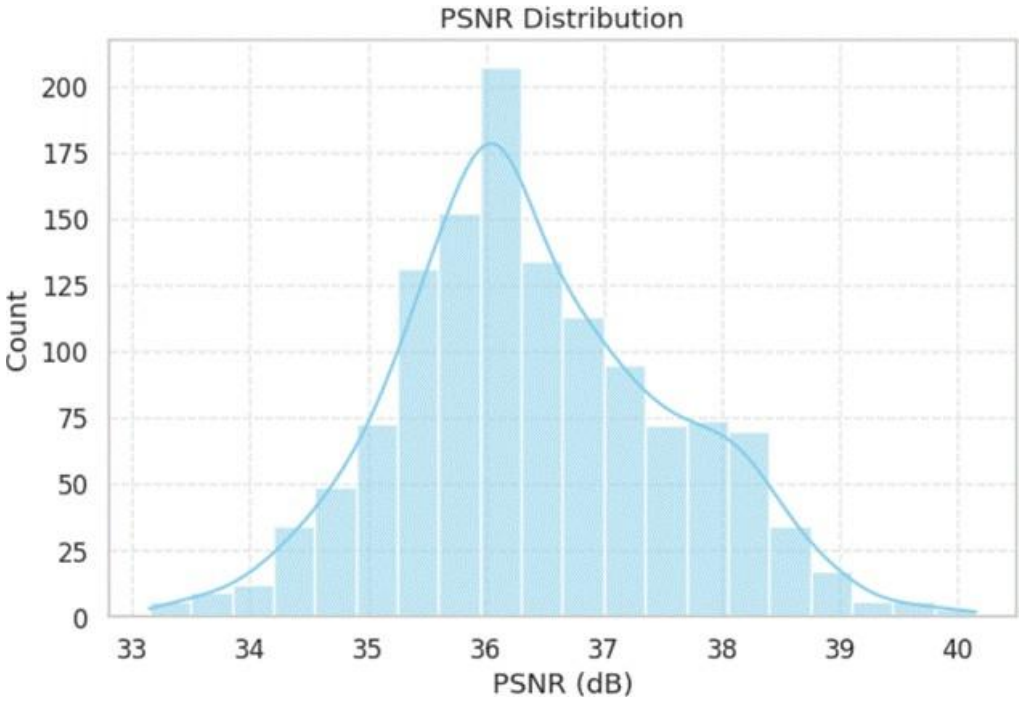
In order to assess the personal contribution of the two major architectural innovations, a variational probabilistic bottleneck and a wavelet-domain PatchGAN discriminator, the ablation study of the architectural component is conducted on the condition of all noise combined. Three variant models are compared: a basic U-Net using wavelet PatchGAN discriminator but no variational bottleneck (Baseline U-Net), U-Net using variational bottleneck but without wavelet decomposition (VAE-GAN No DWT), and the full proposed Dual-Stage VAE-GAN with both. Table 4.3 summarizes the results.

Table 4.3: Architectural Component Ablation Results

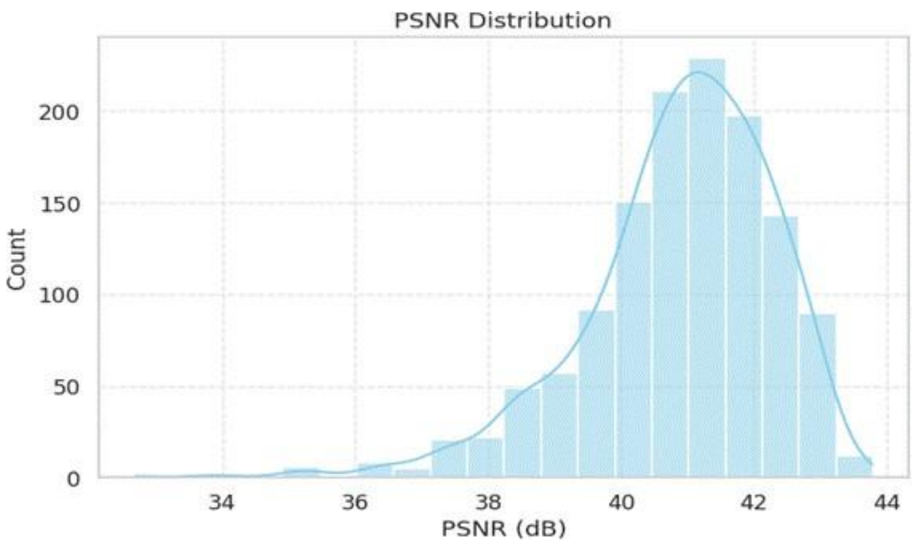
Model Variant	Architecture Description	PSNR (dB)	SSIM
Baseline U-Net	U-Net + Wavelet PatchGAN (No VAE)	32.10	0.9140
No DWT	U-Net + VAE Bottleneck + Image PatchGAN	31.91	0.9087
Proposed Dual-Stage VAE-GAN	**Dual-Stage U-Net + VAE + Wavelet PatchGAN**	**33.50**	**0.9423**

The complementary roles of the two architectural elements are clearly shown by the results of the ablation. The baseline U-Net with no variational bottleneck attains 32.10 dB PSNR and 0.9140 SSIM, which is a significant drop compared to the 37.52 dB PSNR and 0.9624 SSIM of the proposed model. This large performance discrepancy validates that the probabilistic latent modeling offered by the VAE bottleneck is a leading force behind reconstruction quality, which allows the model to model and solve latent uncertainty in the noise-corrupted inputs and not to generate deterministic biased-mean reconstructions. In the absence of the VAE, the lack of its presence causes visible over-smoothing and diffusive error distributions, and residual structured noise artifacts, especially in the homogeneous areas of the brain tissue.

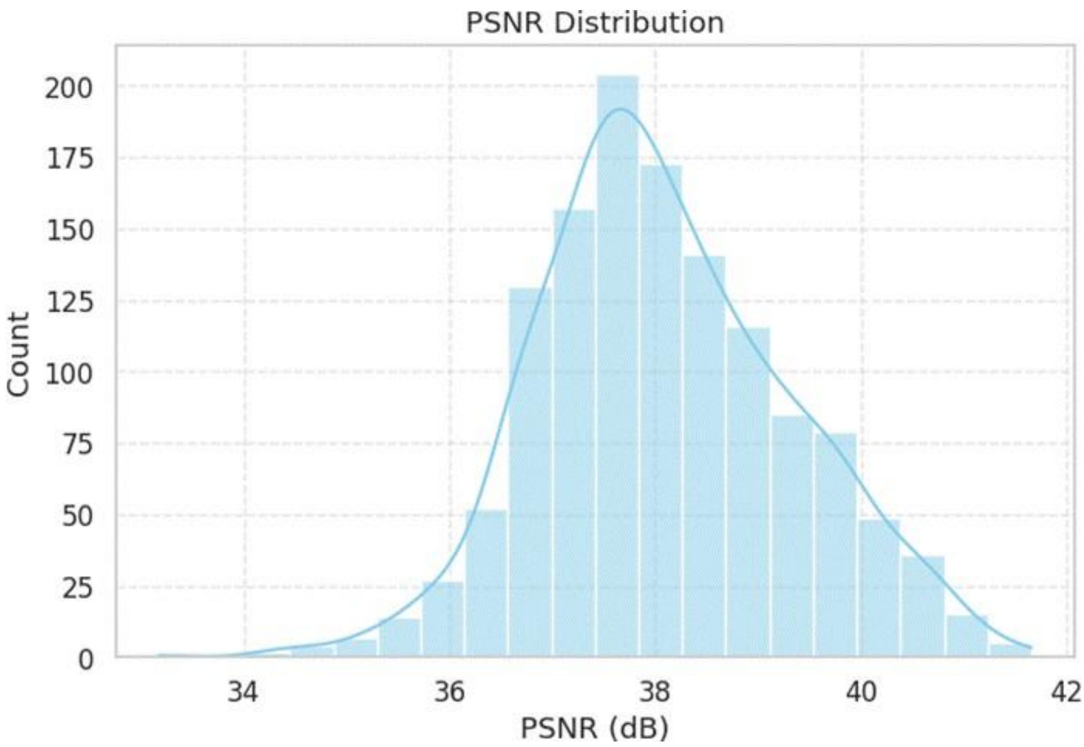
The VAE-GAN without DWT has almost the same performance as the full proposed model (31.91 vs 33.50 dB PSNR; with SSIM improving from 0.9087 to 0.9423) and it means that the bottleneck of the VAE architecture is the most significant architectural factor in structural quality under Rician noise. Nonetheless, qualitative examination indicates that the version without DWT still has unobtrusive residual high-frequency artifact especially the systematic organized noise patterns that are actually suppressed by the wavelet-domain discriminator in the complete model. Gradient energy maps prove that spurious gradient responses in homogeneous regions still exist without DWT, which represents the residue of high-frequency noise that cannot be explicitly identified by the spatial-domain discriminator. These results highlight the special role of the DWT-based discriminator in structured and periodic noise suppression that cannot be well explained by the SSIM metric alone and which is of clinical importance to diagnostic image quality.



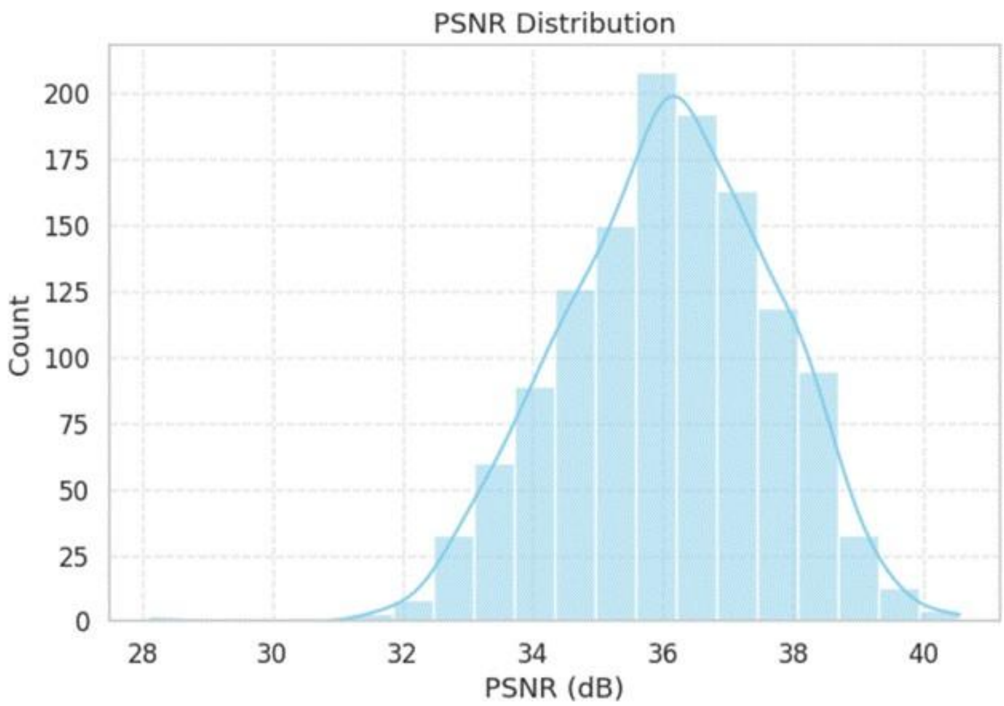
Rician Noise(0.05)



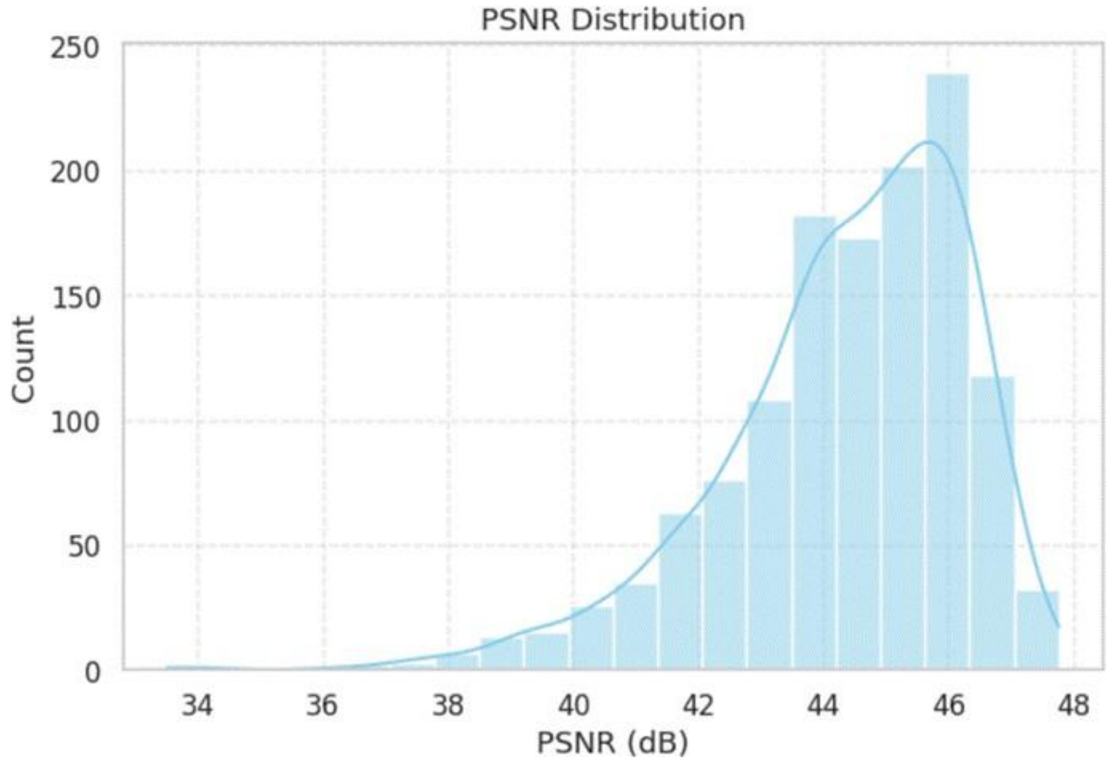
Multiplicative Noise



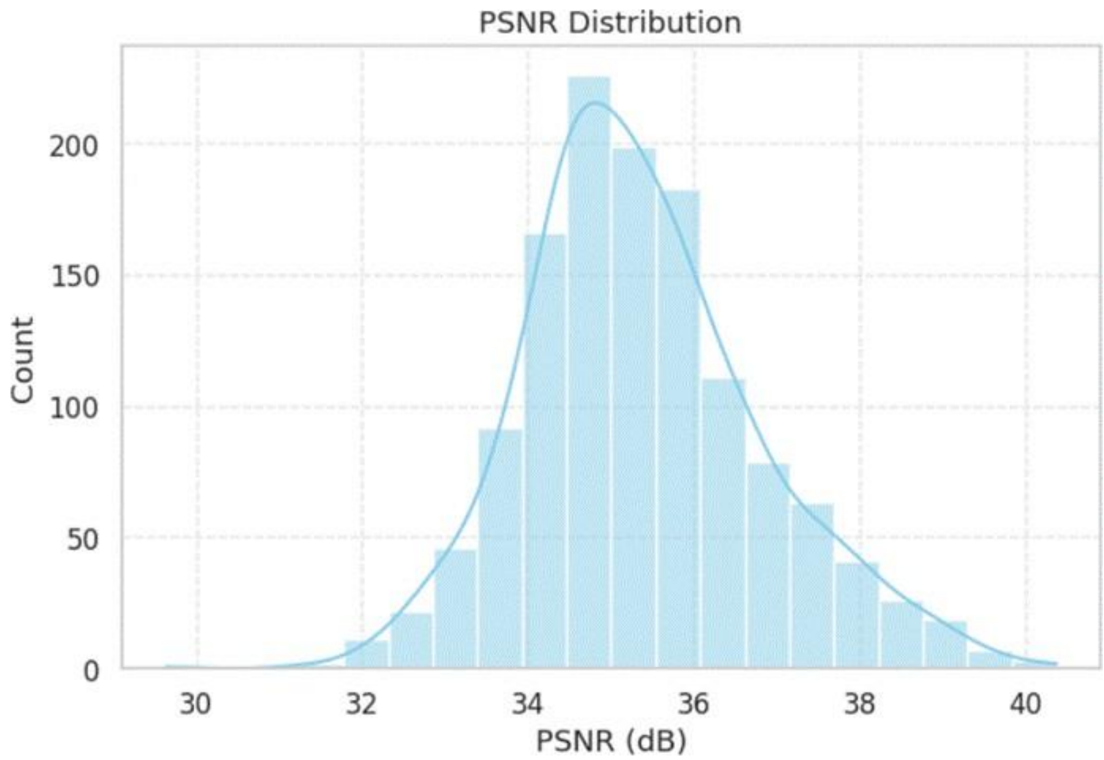
Poisson Noise



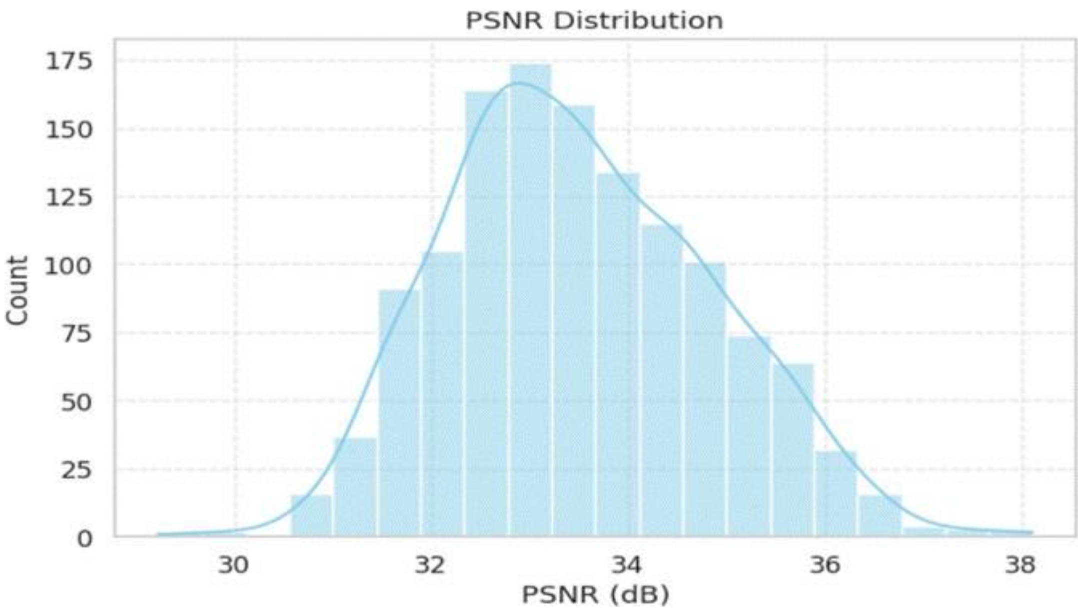
Gaussian Noise



Periodic Noise



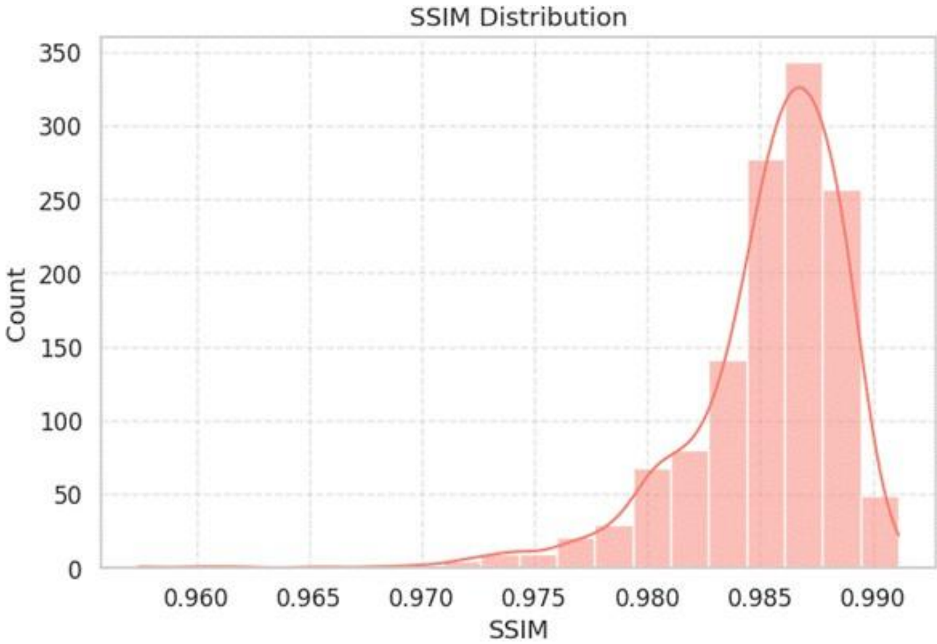
Speckle Noise



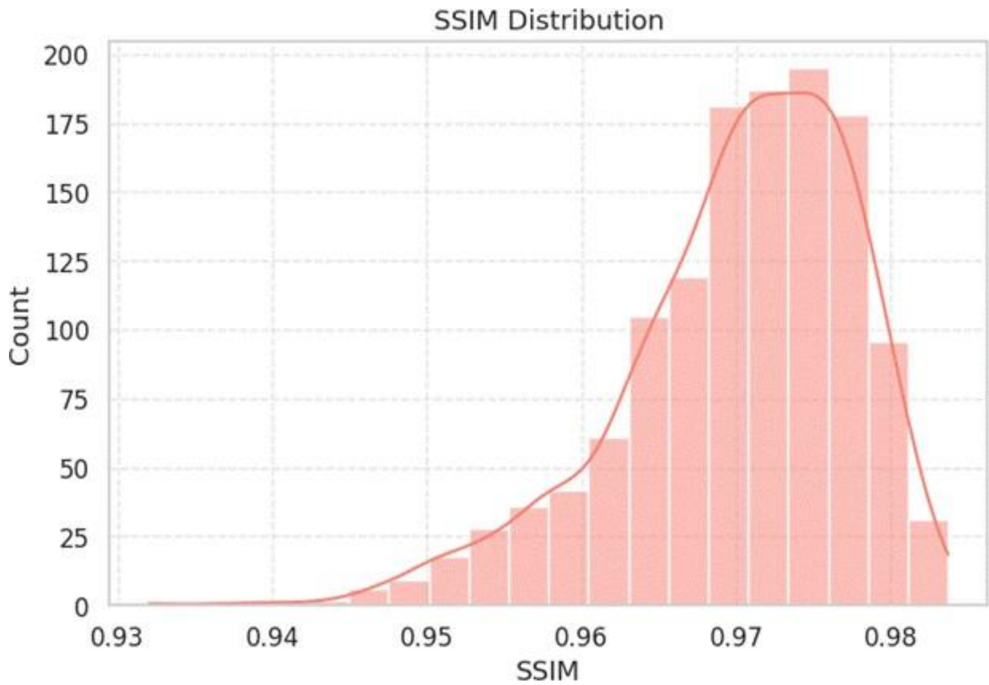
All noise combined

Figure 4.5: PSNR distributions across all noise-type ablation pipelines. Each histogram shows the per-image PSNR spread over the 1,297-image validation set. Poisson noise yields the highest mean PSNR (38.09 dB) while the All-Noise Combined pipeline shows the lowest (33.50 dB)

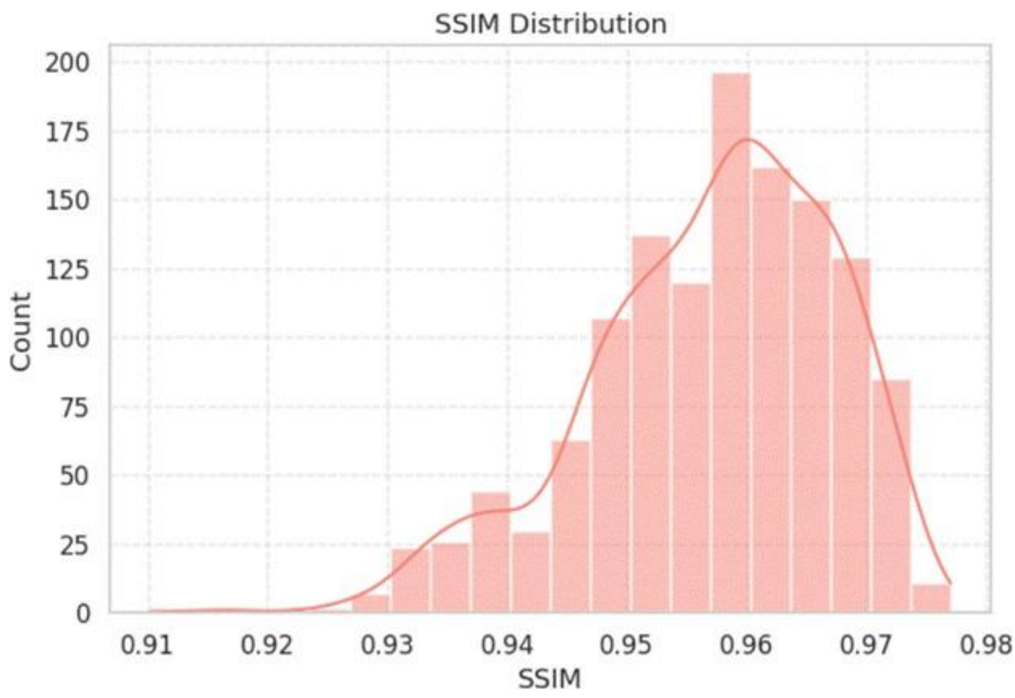
Rician Noise(0.05)



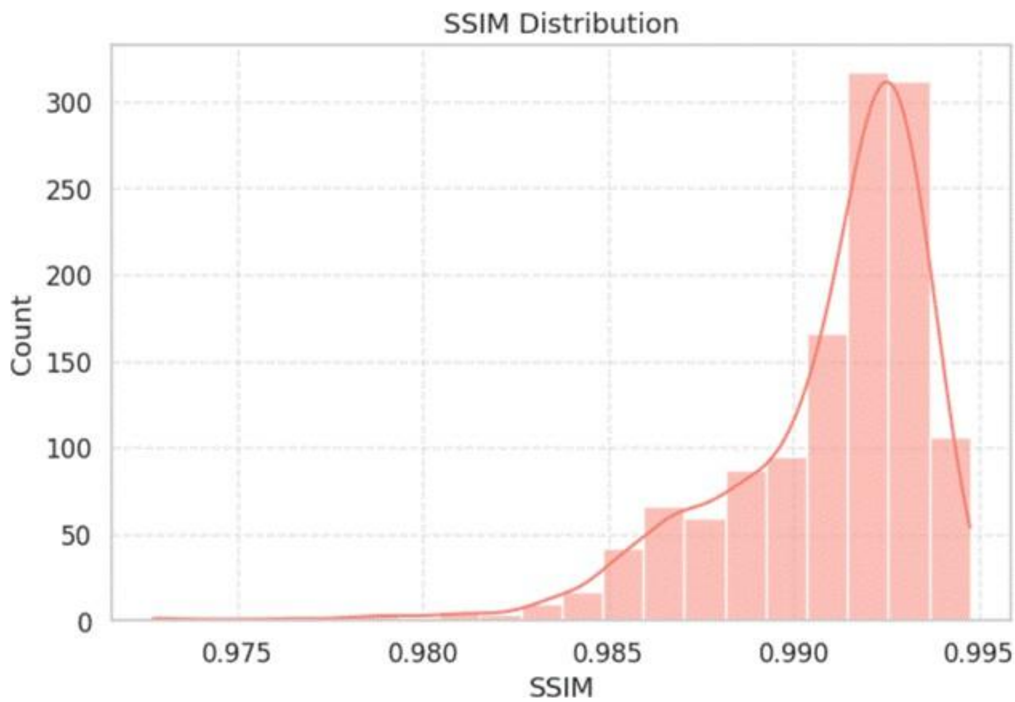
Multiplicative noise



Poisson Noise



Gaussian Noise



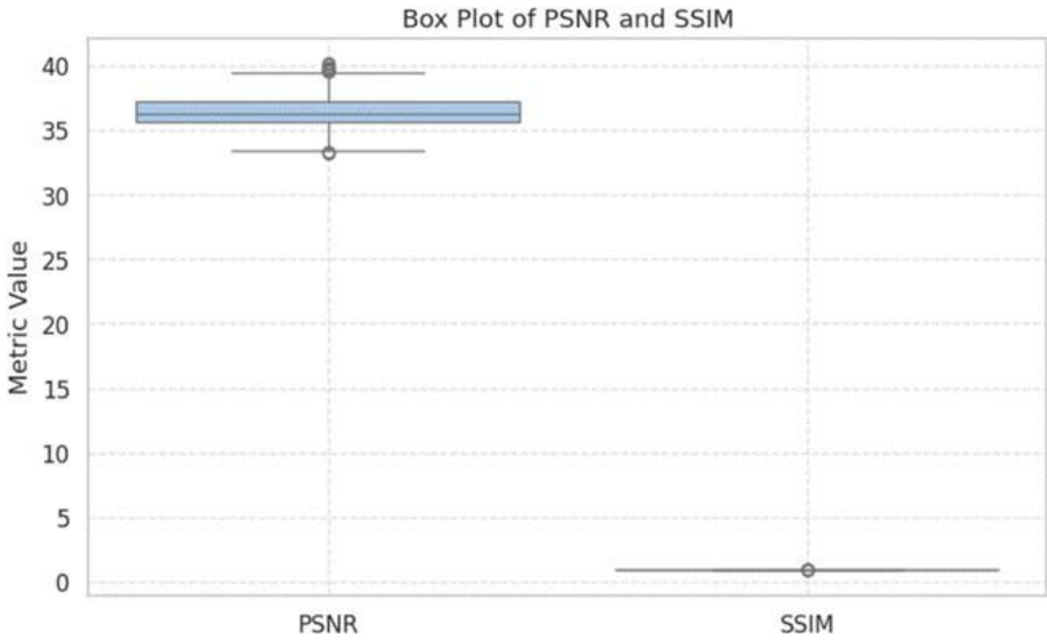
Periodic Noise

Speckle Noise

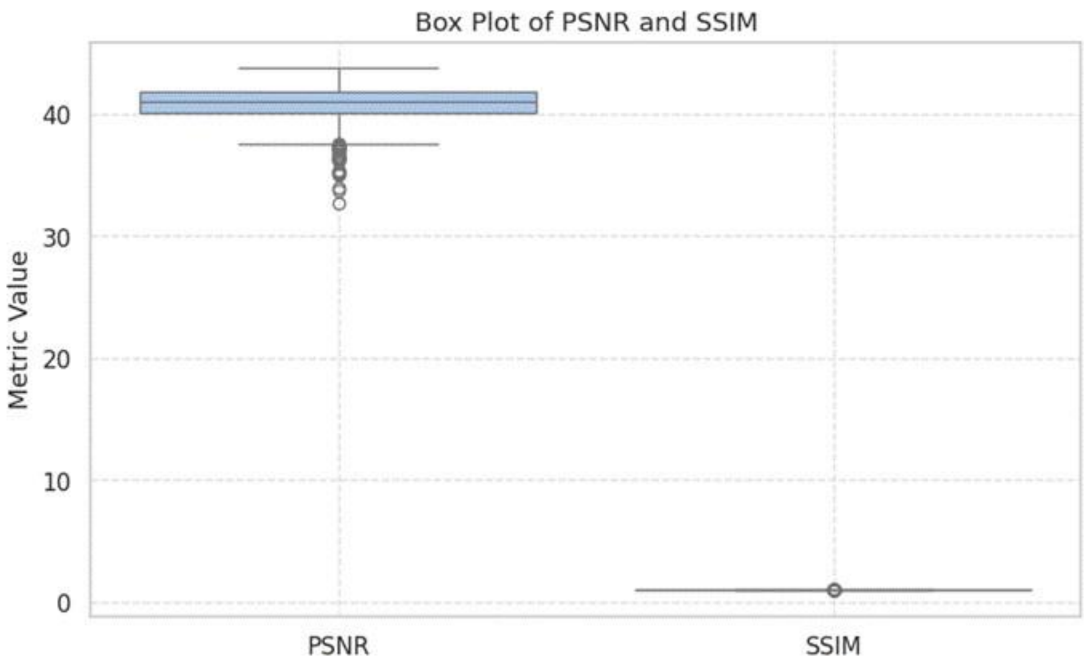
All noise combined

Figure 4.6: SSIM distributions across all noise-type ablation pipelines. Multiplicative noise achieves the highest mean SSIM (0.9853) with the tightest spread (Std = 0.0037), indicating highly consistent structural preservation. The All-Noise Combined pipeline exhibits the widest

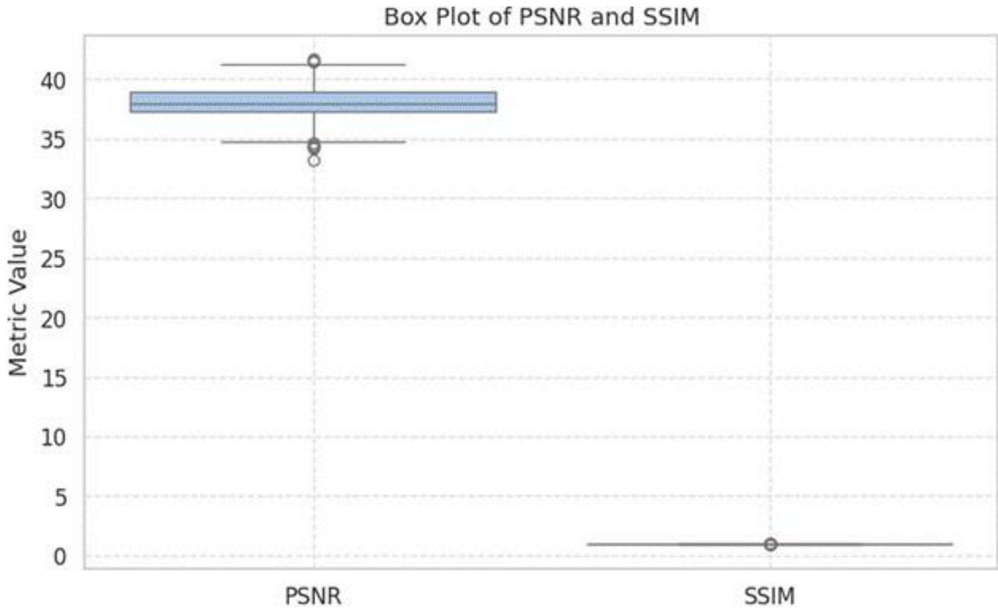
distribution (Std = 0.0129), reflecting increased reconstruction variability under compounded noise.



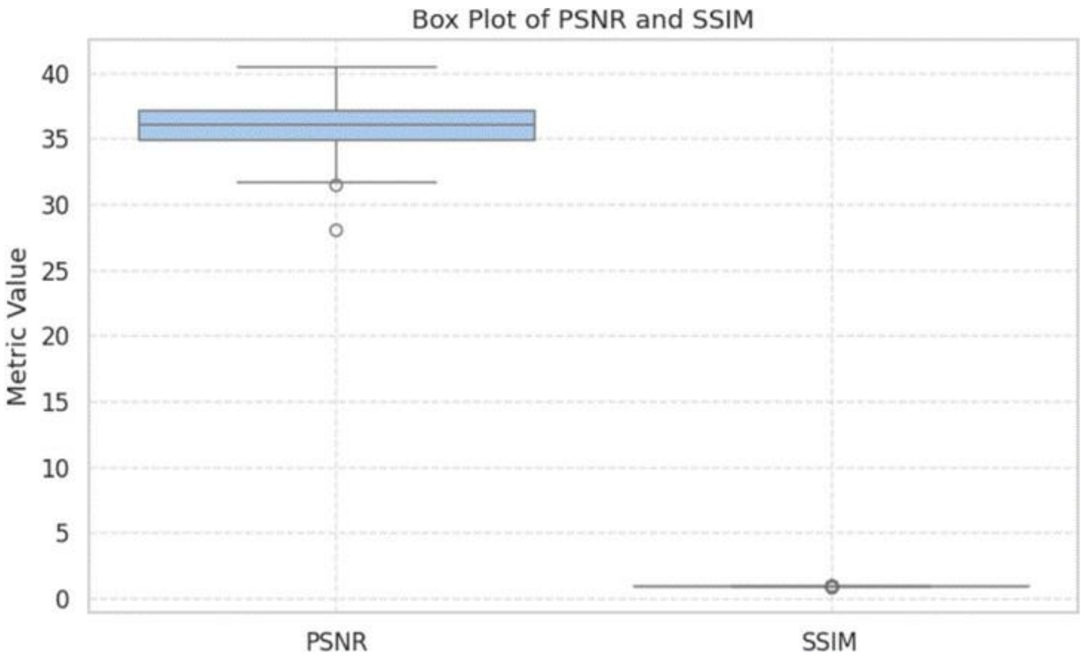
Rician Noise(0.05)



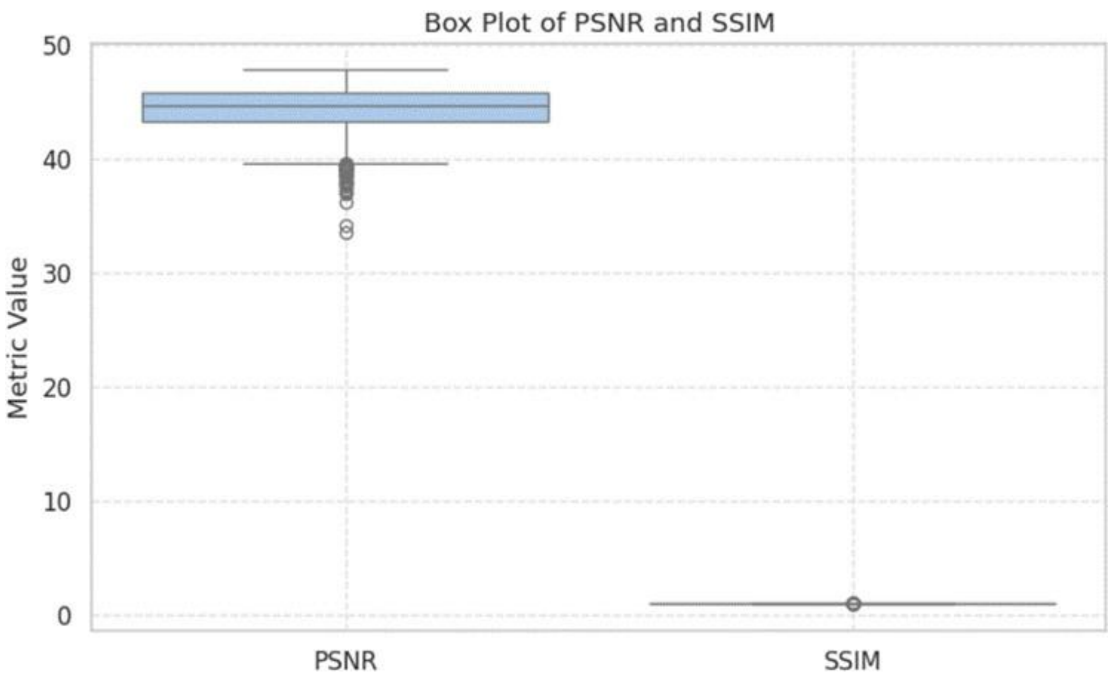
Multiplicative Noise



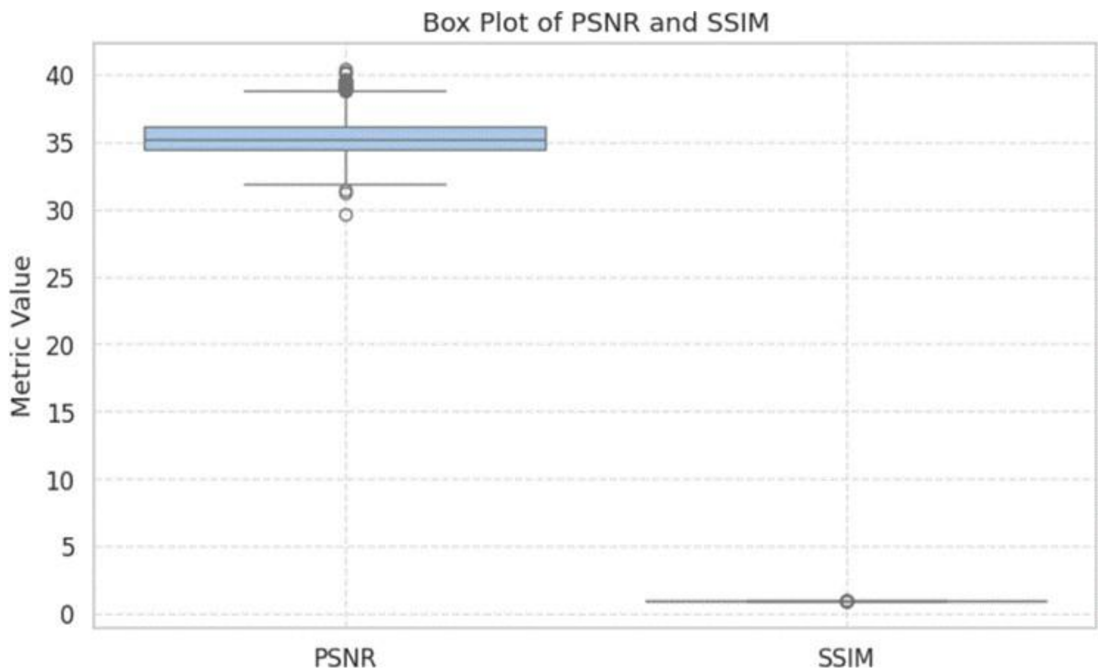
Poisson Noise



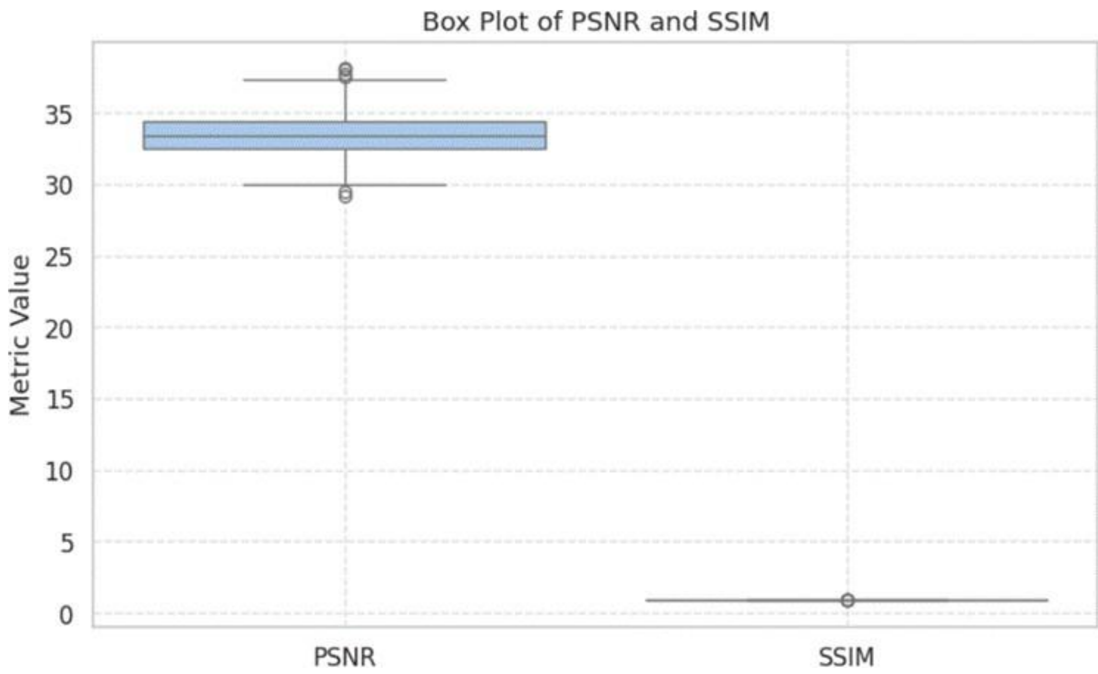
Gaussian Noise



Periodic Noise

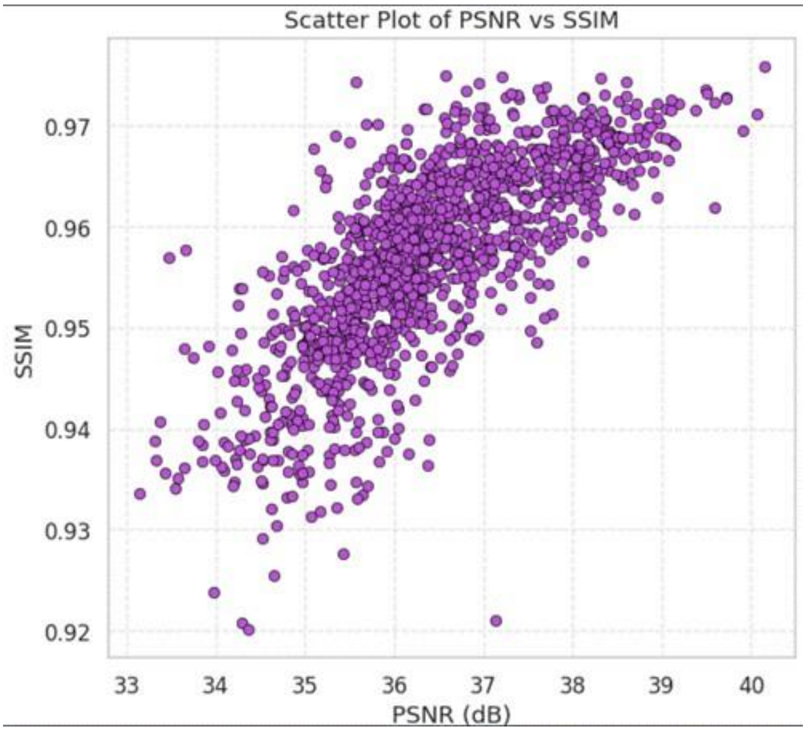


Speckle Noise

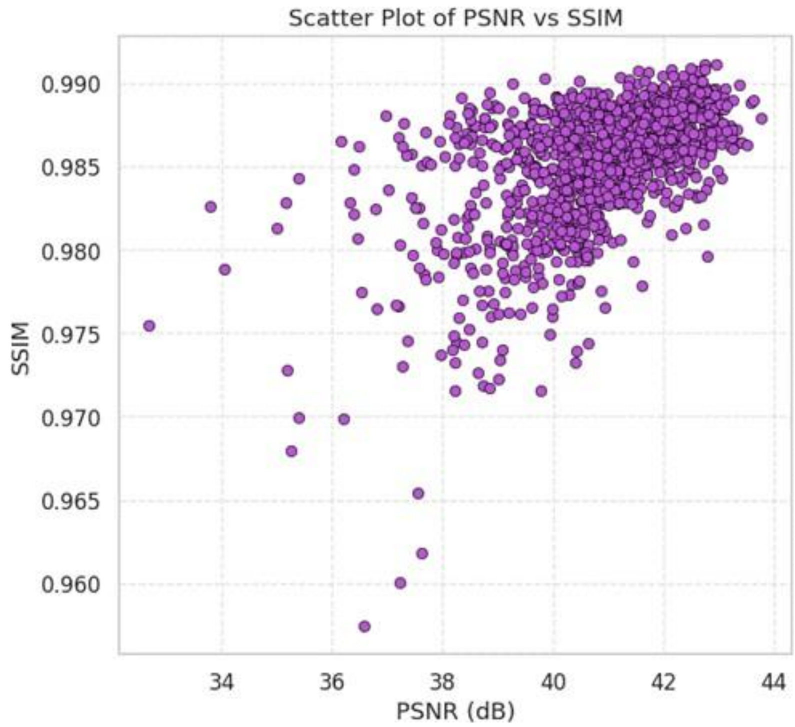


All noise combined

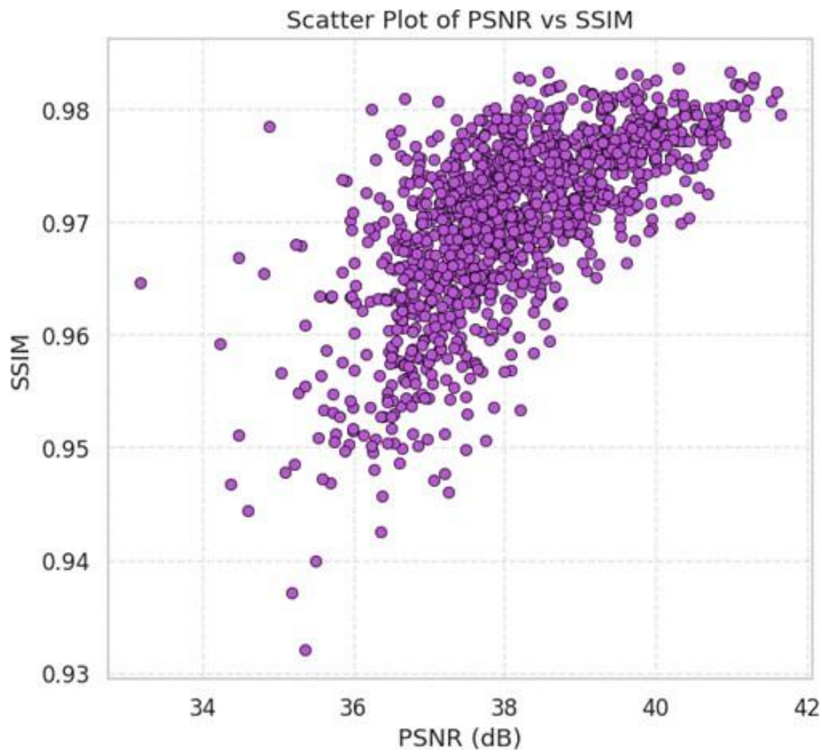
Figure 4.7: Box plots of (a) PSNR and (b) SSIM values across all noise-type ablation pipelines. Each box represents the interquartile range over the validation set, with whiskers extending to the 5th and 95th percentiles. The All-Noise Combined pipeline shows the widest spread on both metrics, while Multiplicative noise produces the most stable SSIM distribution.



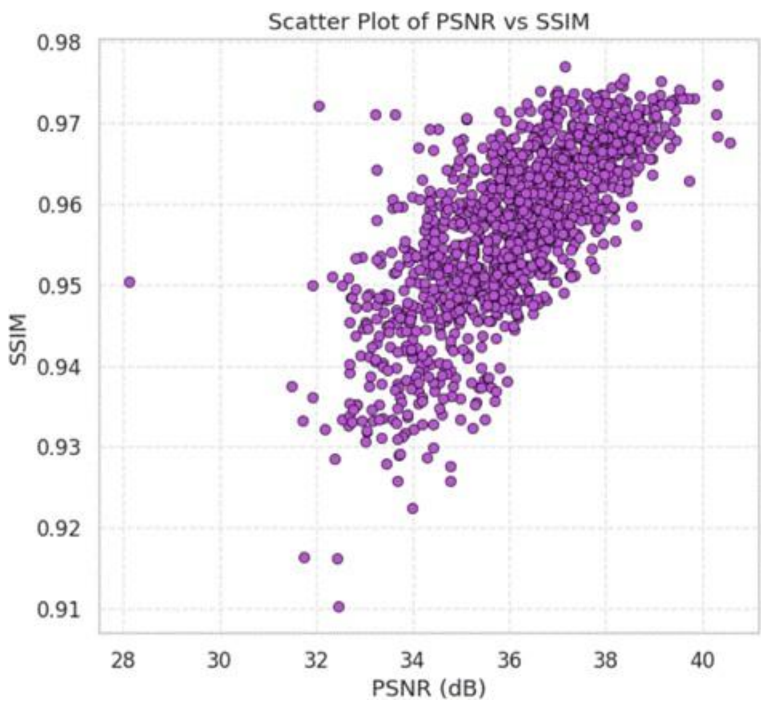
Rician Noise(0.05)



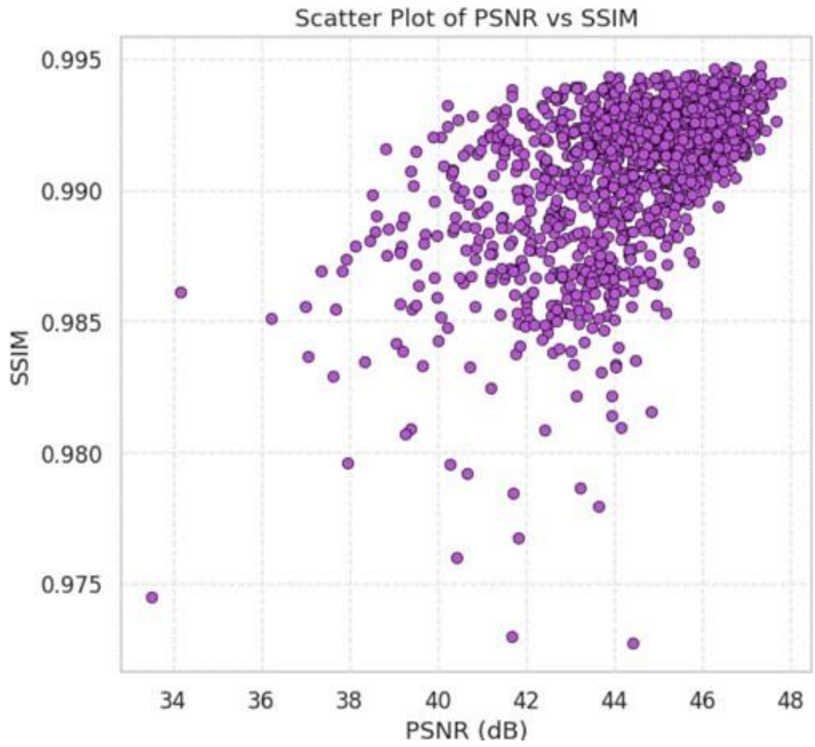
Multiplicative Noise



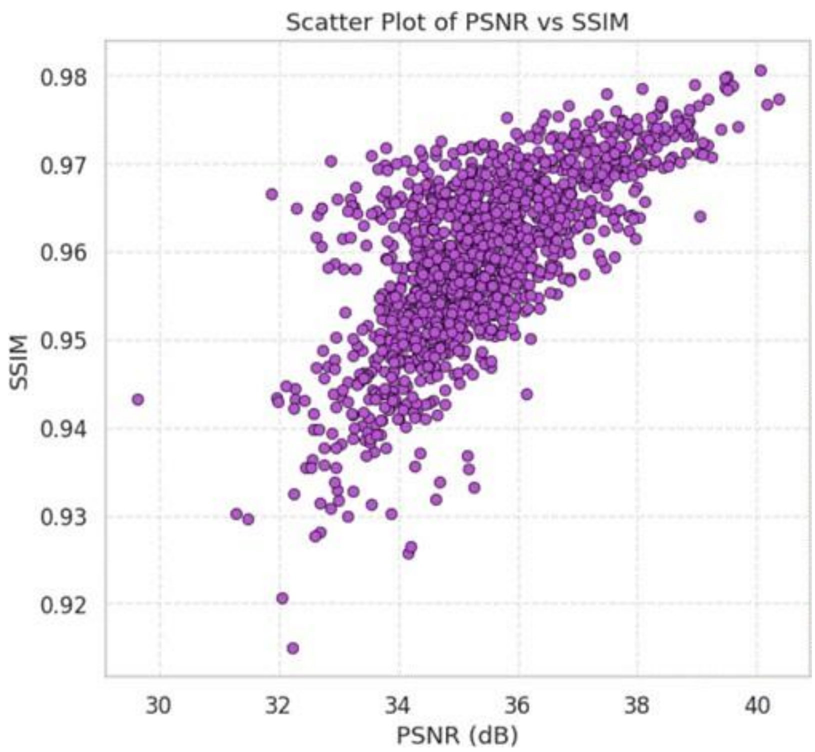
Poisson Noise



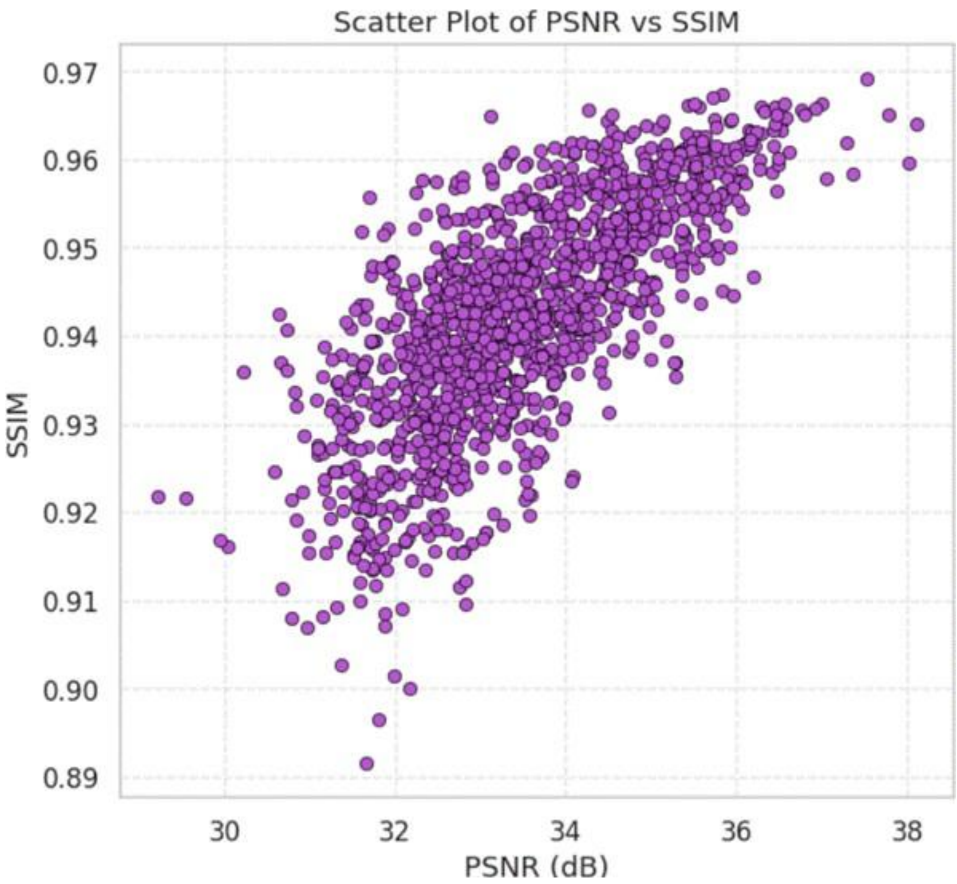
Gaussian Noise



Periodic Noise



Speckle Noise



All Noise combined

Figure 4.8: Scatter plots of per-image PSNR versus SSIM for each noise-type ablation pipeline. Each point corresponds to one validation image. The strong positive correlation observed across all conditions confirms structural consistency in the model's denoising output. The Multiplicative and Poisson pipelines show the tightest clusters at high PSNR and SSIM, while the All-Noise Combined pipeline exhibits the widest dispersion.

CHAPTER 5

Conclusion

5.1 SUMMARY OF WORK

The experimental results clearly show that the proposed Dual-Stage VAE-GAN with the Wavelet PatchGAN discriminator works as a strong and effective MRI denoising model. In comparison to most of the current approaches, which are either specific to one type of noise, or rely on a one-stage model, this model can more efficiently deal with various types of noise. The model can be trained to enhance the pixel quality as well as maintain important anatomical structures by incorporating variational latent modeling, two-stage refinement and frequency-aware adversarial learning, which is highly valued in medical images.

Based on the noise ablation study, it is clear that the model is well generalized when it comes to various types of noise. Periodic noise has the highest performance of 44.34 dB PSNR and 0.9908 SSIM, and multiplicative noise has 40.88 dB and 0.9853. The model still has good performance with 33.50 dB PSNR and 0.9423 SSIM even in the most challenging case when all types of noise are mixed. This demonstrates that the model is rather robust, and the findings are still clinically acceptable even in the presence of excessive noise. This strength is predominantly because of the dual stage design where the former stage eliminates the large noise elements and the latter stage concentrates on finer details that are otherwise overlooked by the single stage models.

Comparing it to other models such as DnCNN and ResUNet, it is evident that traditional CNN-based methods have difficulties with dealing with various types of noise in the first place due to a lack of probabilistic modeling. The proposed model also competes well with SwinIR, which employs transformer-based attention mechanisms, in terms of SSIM, which is more crucial in preserving structure in medical images. It demonstrates that the suggested method can provide high-quality results without the use of complicated and computationally intensive architectures.

In general, the findings indicate that the Dual-Stage VAE-GAN is a stable and scalable model to use in MRI denoising. It offers an adequate tradeoff between image quality, structural preservation, computational efficiency and capability to deal with various forms of noise and consequently is applicable to real world clinical applications.

5.2 KEY FINDINGS

The presented Dual-Stage VAE-GAN model demonstrates good results in the MRI denoising of various and complicated noise cases. It achieves a mean PSNR of 33.50 dB and SSIM of 0.9423 with vendor-specific noise, which is better than the baseline models such as DnCNN, ResUNet and SwinIR. The increased values of SSIM are used to show a better preservation of anatomical structures, which is essential in clinical interpretation.

The model is able to deal with heterogeneous and mixed noise distributions with structural fidelity even in more difficult cases. This performance can be credited to the dual-stage design that first stage does coarse denoising and restores global structure and the second stage refines the result by eliminating residual noise and sharpening fine detail.

A wavelet-domain PatchGAN discriminator is also integrated to enhance the performance as it allows frequency-sensitive learning which contributes to saving the low-frequency structure and high-frequency details. This is especially useful in dealing with periodic and structured noise.

Also, the model has shown healthy performance in various types of noise, and varying Rician noise levels, whereby behavior reduces gradually with increase in noise. The framework is highly structurally similar even under the all-noise combined scenario, which proves its strength and applicability to real-life clinical settings.

5.3 LIMITATIONS

The proposed framework has a number of limitations that should be noted. Although the proposed model achieves better results than all baselines on the vendor-aware pipeline the results are only evaluated on a single glioma MRI dataset and at a given noise severity regime. It should be mentioned that the performance difference between SwinIR and PSNR (33.50 vs 32.87 dB) is smaller than the significantly larger SSIM difference indicating that transformer-based attention can be more easily scaled with more varied training data. Future research may consider the incorporation of self-attention in the dual-stage VAE to incorporate long-range dependency modelling in addition to frequency-aware adversarial supervision that already exists in this architecture.

Second, the computational complexity of the dual-stage variational architecture is in and of itself greater than the single-stage models. Both independent variational encoder-decoder modules are trained on 256x256 inputs with a 131,072 dimensional flattened bottleneck, which results in the heavy memory demands during training. The batch sizes are thus restricted to 2 and this can cause certain noise on the gradient estimation.

Third, the present assessment is performed only on a glioma MRI data of one imaging modality and anatomy. The applicability of the proposed framework to other types of MRI (e.g. T1w, FLAIR), anatomical areas, or real clinical noise distributions without synthetic simulation has not been empirically tested and is a valuable avenue of future clinical implementation.

Fourth, the synthetic noise simulation pipelines, though clinically motivated, are approximations of actual scanner-specific noise properties. Clinical MRI noise is in reality dependent on field strength, coil arrangement, pulse sequence factors and patient motion, which cannot be completely modeled by the noise generation procedures of fixed parameters used in the present work. It would be valuable in the future to extend the evaluation to actual acquisitions of physical MRI scanners or emulator frameworks of scanners.

5.4 FUTURE WORK

The work may be extended and improved in a number of ways. The first direction is to transition away to 3D MRI volumes, which can be used to obtain a more detailed spatial representation and increase the overall quality of the denoising. Additionally, paired datasets are not the only method to be considered; self-supervised or unsupervised methods can be investigated in such a way that the model could be trained on real-life data where clean reference images are not always present.

The other helpful measure would be testing the model on real clinical noisy data so that it is easier to understand how the model works in a real-life scenario. The model is also versatile because it can be applied to other types of imaging, such as CT or PET. Moreover, it is possible to create light versions of the model to decrease the cost of computation and make this model applicable in clinical practice in real-time.

Other refinements can be made by incorporating adaptive noise modeling such that the model is made to better deal with unknown or varying types of noise. Adding attention mechanisms or transformer-based elements could also contribute to the enhancement of performance. Lastly, radiologists can be validated on the model and domain adaptation methods can be used to make sure that the model is reliable in other hospitals and scanner environments.

Appendix

OVERVIEW

The given appendix contains some technical information concerning the implementation of the proposed Dual-Stage VAE-GAN architecture to perform MRI image denoising. It contains the supporting information which is not entirely discussed in the principal report but is significant to have a clearer idea of the whole system.

Most of the information in the appendix is related to the model architecture, noise generation process, training procedure, and the important implementation steps. It also contains some of the selected code snippets to provide a clear picture of the implementation of the proposed method in practice.

Furthermore, there are also additional findings and assessment information that serves to verify the analysis of performance found in the rest of the chapters. In general, this part assists in making this work in this project more complete and realistic.

GENERATOR MODEL IMPLEMENTATION

The following code shows the implementation of the proposed Dual-Stage U-Net based VAE Generator used for MRI image denoising

```
class DualStageUNetGenerator(nn.Module):
    def __init__(self, in_channels=1, out_channels=1, latent_dim=256):
        super().__init__()


        def conv_block(in_ch, out_ch):
            return nn.Sequential(
                nn.Conv2d(in_ch, out_ch, 3, 1, 1),
                nn.InstanceNorm2d(out_ch),
                nn.LeakyReLU(0.2, inplace=True)
 # === Stage 1: Encoder ===
        self.enc1 = conv_block(in_channels, 64)
```

```
        self.enc2 = conv_block(64, 128)
        self.enc3 = conv_block(128, 256)
        self.enc4 = conv_block(256, 512)
        self.pool      = nn.MaxPool2d(2)
        self.flatten   = nn.Flatten()
        self.unflatten = nn.Unflatten(1, (512, 16, 16))

        self.pre_vae_norm1  = nn.LayerNorm(512 * 16 * 16)
        self.pre_vae_norm2  = nn.LayerNorm(512 * 16 * 16)
        self.post_dec_norm1 = nn.LayerNorm(512 * 16 * 16)
        self.post_dec_norm2 = nn.LayerNorm(512 * 16 * 16)

# === Stage 1: Coarse VAE ===
        self.fc_mu1     = nn.Linear(512 * 16 * 16, latent_dim)
        self.fc_logvar1 = nn.Linear(512 * 16 * 16, latent_dim)
        self.fc_dec1    = nn.Linear(latent_dim, 512 * 16 * 16)

        self.up1_s1  = nn.ConvTranspose2d(512, 256, 2, 2)
        self.dec1_s1 = conv_block(512, 256)
        self.up2_s1  = nn.ConvTranspose2d(256, 128, 2, 2)
        self.dec2_s1 = conv_block(256, 128)
        self.up3_s1  = nn.ConvTranspose2d(128, 64, 2, 2)
        self.dec3_s1 = conv_block(128, 64)
        self.out_s1  = nn.Conv2d(64, out_channels, 1)

        # === Stage 2: Fine Refinement VAE ====
```

```
        self.enc1_s2 = conv_block(in_channels + out_channels, 64)
        self.enc2_s2 = conv_block(64, 128)
        self.enc3_s2 = conv_block(128, 256)
        self.enc4_s2 = conv_block(256, 512)

        self.fc_mu2    = nn.Linear(512 * 16 * 16, latent_dim)
        self.fc_logvar2 = nn.Linear(512 * 16 * 16, latent_dim)
        self.fc_dec2    = nn.Linear(latent_dim, 512 * 16 * 16)

        self.up1_s2  = nn.ConvTranspose2d(512, 256, 2, 2)
        self.dec1_s2 = conv_block(512, 256)
        self.up2_s2  = nn.ConvTranspose2d(256, 128, 2, 2)
        self.dec2_s2 = conv_block(256, 128)
        self.up3_s2  = nn.ConvTranspose2d(128, 64, 2, 2)
        self.dec3_s2 = conv_block(128, 64)
        self.out_s2  = nn.Conv2d(64, out_channels, 1)

    def reparameterize(self, mu, logvar):
        logvar = torch.clamp(logvar, min=-4.0, max=4.0)
        std    = torch.exp(0.5 * logvar)
        eps    = torch.randn_like(std)
        return mu + eps * std, logvar

    def forward(self, x):
# ── Stage 1 Encoder ──
        e1 = self.enc1(x)
```

```
e2 = self.enc2(self.pool(e1))
e3 = self.enc3(self.pool(e2))
e4 = self.enc4(self.pool(e3))

flat       = self.pre_vae_norm1(self.flatten(self.pool(e4)))
mu1        = self.fc_mu1(flat)
logvar1_raw = self.fc_logvar1(flat)
z1, logvar1 = self.reparameterize(mu1, logvar1_raw)

z1 = self.unflatten(self.post_dec_norm1(self.fc_dec1(z1)))
z1 = F.interpolate(z1, size=(32, 32), mode='bilinear', align_corners=False)

d1 = self.up1_s1(z1)
d1 = self.dec1_s1(torch.cat([d1, e3], dim=1))
d2 = self.up2_s1(d1)
d2 = self.dec2_s1(torch.cat([d2, e2], dim=1))
d3 = self.up3_s1(d2)
d3 = self.dec3_s1(torch.cat([d3, e1], dim=1))
x1 = torch.sigmoid(self.out_s1(d3))   # Coarse output

# ── Stage 2 Encoder ──
stage2_in = torch.cat([x1, x], dim=1)
e1_s2 = self.enc1_s2(stage2_in)
e2_s2 = self.enc2_s2(self.pool(e1_s2))
e3_s2 = self.enc3_s2(self.pool(e2_s2))
e4_s2 = self.enc4_s2(self.pool(e3_s2))
```

```
flat_s2     = self.pre_vae_norm2(self.flatten(self.pool(e4_s2)))
mu2         = self.fc_mu2(flat_s2)
logvar2_raw = self.fc_logvar2(flat_s2)
z2, logvar2 = self.reparameterize(mu2, logvar2_raw)

z2 = self.unflatten(self.post_dec_norm2(self.fc_dec2(z2)))
z2 = F.interpolate(z2, size=(32, 32), mode='bilinear', align_corners=False)

d1 = self.up1_s2(z2)
d1 = self.dec1_s2(torch.cat([d1, e3_s2], dim=1))
d2 = self.up2_s2(d1)
d2 = self.dec2_s2(torch.cat([d2, e2_s2], dim=1))
d3 = self.up3_s2(d2)
d3 = self.dec3_s2(torch.cat([d3, e1_s2], dim=1))
x2 = torch.sigmoid(self.out_s2(d3))   # Fine output

return x2, x1, mu1, logvar1, mu2, logvar2
```

This code is the application of the suggested Dual-Stage U-Net based VAE Generator. The model uses two steps of MRI image denoising to enhance noise removal and preservation of structure.

The noisy input image is then processed by several layers of convolutional layers in the first step to extract features. Such features are then transformed into a latent representation by the VAE mechanism, the mean and variance are learned and sampling is performed with the reparameterization trick. This phase generates a low-quality denoised result.

The second stage involves using the coarse output and adding it to the original input image and then the encoder is used again. This assists the model to hone the output by eliminating the left-over noise and enhancing finer details.

In general, this architecture enables gradual denoising, in which the initial step removes the biggest noise and the second step improves the image quality further. This improves performance especially in complex and mixed noise conditions.

DISCRIMINATOR AND WAVELET TRANSFORM

The purpose of this appendix is to provide an overview of how to create a discriminative network for the Dual-Stage VAE-GAN framework, which will serve as a way to enhance the quality of MRI images generated via VAE-GANs by providing a method for distinguishing real (clean) MRI images and denoised MRI image outputs from the generator.

One of the key features of this implementation involves using the 2D Discrete Wavelet Transform (DWT) as an input layer to the Discriminator's network. By using DWT, the model is able to derive frequency information from the input images, allowing the model to make better distinctions between the structural vs. textural characteristics of the input images. In addition, DWT increases the Discriminator's sensitivity to noise and small variations in the input images by operating within the wavelet domain.

```
class DWT_2D(nn.Module):
    def _init_(self):
        super(DWT_2D, self).__init__()
        h  = torch.tensor([1.0,  1.0]) / (2 ** 0.5)
        g  = torch.tensor([1.0, -1.0]) / (2 ** 0.5)
        ll =  torch.ger(h,  h).unsqueeze(0).unsqueeze(0)
        lh = torch.ger(g,  h).unsqueeze(0).unsqueeze(0)
        hl = torch.ger(h,  g).unsqueeze(0).unsqueeze(0)
        hh = torch.ger(g, g).unsqueeze(0).unsqueeze(0)
        self.register_buffer('filters', torch.cat([ll, lh, hl, hh], dim=0))

    def forward(self, x):
        f = self.filters.to(dtype=x.dtype)
        return F.conv2d(x, f, stride=2, padding=0)

class Discriminator(nn.Module):
    def __init__(self, in_channels=1):
```

```
        super(Discriminator, self)._init_()
        self.dwt = DWT_2D()
        self.net = nn.Sequential(
            nn.Conv2d(4,  64,  kernel_size=4, stride=2, padding=1),
            nn.LeakyReLU(0.2, inplace=True),
            nn.Conv2d(64,  128, kernel_size=4, stride=2, padding=1),
            nn.InstanceNorm2d(128),
            nn.LeakyReLU(0.2, inplace=True),
            nn.Conv2d(128, 256, kernel_size=4, stride=2, padding=1),
            nn.InstanceNorm2d(256),
            nn.LeakyReLU(0.2, inplace=True),
            nn.Conv2d(256, 512, kernel_size=4, stride=2, padding=1),
            nn.InstanceNorm2d(512),
            nn.LeakyReLU(0.2, inplace=True),
            nn.Conv2d(512, 1,   kernel_size=4, stride=1, padding=1)
        )

    def forward(self, x):
        x = self.dwt(x)
        return self.net(x)
```

In this wavelet transform, the input is first broken down into several frequencies that are then fed into convolution layers. The use of convolution layers will allow the model to understand essential features regarding image quality, noise, and consistency.

Use of a PatchGAN architecture helps to focus on the local area of the image rather than analyzing the whole picture. It enables the model to capture small changes or distortions in the image and preserve fine details in the resulting denoised image.

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